

Windows
Manual

Learning and Behavior Laboratory

NSC3120/PSY3120



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Baylor University

**Department of Psychology and
Neuroscience**

Student, TA, Lab Partner, and Animal Information

Your Information:	Your Laboratory Partner's Information:
Name: _____	My Lab Partner's Name is: _____
My Lab Sessions occur on: Mon/Wed Tue/Thu	Their Lab Sessions occur on: Mon/Wed Tue/Thu
My Lab Sessions start at: _____	Their Lab Session starts at: _____
My Lab Instructor is : _____	My Lab Partner's contact information is:
My lab TA's cell phone number is: _____	Email: _____
	Phone: _____
	Their Lab Instructor is : _____
	Their lab's instructor's cell phone number is: _____

Rat Information

Your Rat's Name: _____

Your Rat's Cage Number: _____

Your Rat's Initial Weight (in grams): _____

80% of your Rat's Initial Weight (in grams): _____

Learning and Behavior Lab: Schedule of Events

Date	Day	Lab Week	Event	Quizzes/Exams/ Assignment(s) Due
8/26/2025	Tues		Class Introduction Sniffy™ The Virtual Rat	
8/28/2025	Thurs		Sniffy™ The Virtual Rat	No in-person class
8/29/2025	Friday		Sniffy™ The Virtual Rat	Sniffy Projects 1, 2, 3
9/2/2025	Tues		Sniffy™ The Virtual Rat	No in-person class
9/4/2025	Thurs		Sniffy Exam (administered during your scheduled lab session, In- person)	Sniffy Exam; Sniffy Projects 4, 5, 6, 7
9/9/2025	Tues		Experiment 1: Handling	Quiz 1
9/11/2025	Thurs		Experiment 2: Magazine Training	Quiz 2
9/16/2025	Tues		Experiment 3: Shaping	Quiz 3
9/18/2025	Thurs		Experiment 3: Shaping	
9/23/2025	Tues		Experiment 4: Continuous Reinforcement	Quiz 4
9/25/2025	Thurs		Experiment 4: Continuous Reinforcement	
9/30/2025	Tues		Experiment 5: Extinction & Recovery	Quiz 5
10/2/2025	Thurs		Experiment 5: Extinction & Recovery	
10/3/2025	Fri			Assignment 1
10/7/2025	Tues		Experiment 6: Partial Reinforcement	Quiz 6
10/9/2025	Thurs		Experiment 6: Partial Reinforcement	
10/14/2025	Tues		Experiment 7: Partial Reinforcement Extinction Effect	Quiz 7
10/16/2025	Thurs		Experiment 7: Partial Reinforcement Extinction Effect	
10/17/2025	Fri			Assignment 2
10/21/2025	Tues		Experiment 8: Priming Event	
10/23/2025	Thurs		Experiment 8: Priming Event	

10/24/2025	Fri			RO Prospectus
10/28/2025	Tues		Experiment 8: Choice Behavior	Quiz 8
10/30/2025	Thurs		Experiment 8: Choice Behavior	
10/31/2025	Fri		Boxes for Rat Olympics MUST Be Turned in By This Date	Assignment 3
11/4/2025	Tues		Experiment 9: Stimulus Discrimination	Quiz 9
11/6/2025	Thurs		Experiment 9: Stimulus Discrimination	
11/7/2025	Fri			RO Preliminary Write-up
11/11/2025	Tues		Rat Olympics (RO) Training	
11/13/2025	Thurs		Rat Olympics (RO) Training	
11/14/2025	Fri			Assignment 4
11/18/2025	Tues		Rat Olympics (RO) Training	
11/20/2025	Thurs		Rat Olympics (RO) Training	
11/21/2025	Fri			Assignment 5
11/25/2025	Tues		Thanksgiving Break	
11/27/2025	Thurs		Thanksgiving Break	
12/2/2025	Tues		Rat Olympics (RO) Training	
12/4/2025	Thurs		Rat Olympics (RO) Demonstration	
12/9/2025	Tues		Learning & Behavior Lab Final Exam (administered during your scheduled lab session, In-person)	Final Exam
12/10/2025	Wed			RO Final Report

**Contacts for
Emergency, Weekend, Holiday Care Issues**

Name	Position	Telephone number
For an emergency situation		710-2222
Dr. Joaquin Lugo	PI	710-2389
Dr. Joaquin Lugo	Protocol Contact	710-2389
Shanika Avila	Manager	254-709-9911

**Any Questions or Concerns about Animal Care & Use,
Animal Welfare or Human Safety?**

For animal welfare questions, concerns, or complaints:

- BU veterinarian – Jon Bova, 254-710-2541 or Jon_Bova@Baylor.edu
- Institutional Animal Care and Use Committee – 254-710-1438 or IACUC@baylor.edu

For human health and safety concerns:

- Environmental Health and Safety Office – 254-710-1031 or Dennis_Nolan@baylor.edu

Anonymous Reporting

Anonymous Reporting can be made 24/7 on EthicsPoint at
<https://baylor.edu/ethicspoint> or
 866-384-4277 (toll-free)

No one shall be discriminated against or be subject to any reprisal for reporting violations or deficiencies in animal care and use and/or scientific misconduct.

NSC/PSY3120 Learning & Behavior Lab Syllabus, Policies, and Procedures

Relationship between the Learning & Behavior Lab (NSC/PSY3120) and the Lecture Course (NSC/PSY3320):

This lab has been designed to help you better understand learning & behavior phenomena, so it should help your performance in the lecture course (NSC/PSY3320). However, your grade in this lab is separate from your grade in the course (*see below for the lab's grading criteria*). Concurrent enrollment in the lecture and lab is preferred but not essential.

Relationship between you, the Learning & Behavior Lab (NSC/PSY3120) Teaching Assistants (TAs), and the professor(s) in the Learning & Behavior Course(s) (NSC/PSY3320): The Teaching Assistants (TA's) who run the learning & behavior lab sessions are responsible for their administration and for all day-to-day activities, so you should first talk to them about any lab-related issues or problems that you might be experiencing. However, if you don't get a satisfactory answer from them, you should then bring the problem to the attention of the professor(s) who conduct the Learning & Behavior lecture course(s).

Goals of the Learning & Behavior Laboratory: As previously stated, the NSC/PSY3120 lab is designed to expand your understanding of learning phenomena. More specifically, these Sniffy projects and the operant conditioning experiments have been designed to illustrate the general principles of learning and behavior theory. By enrolling in this lab, you are given the opportunity to conduct a series of learning experiments on both a virtual rat and with a live rat to investigate numerous learning phenomena. More specifically, you will be given the opportunity to:

- Work independently and in a pair group
- To test and apply several aspects of non-associative, Pavlovian, and operant conditioning
- Observe, record, and quantify "investigatory" and foraging behavior
- Magazine train a live rat in an operant conditioning chamber
- Shape a lever-pressing response in a live rat in an operant conditioning chamber, henceforth often referred to as a "Skinner box" in this lab manual.
- Quantify and evaluate this response under continuous reinforcement (CRF).
- Partially extinguish and then recover this response.
- Explore the partial reinforcement extinction effect.
- Investigate *choice behavior*.
- Affect a degree of behavioral control over your rat via stimulus discrimination.

Semester Sequence of Events:

- **Phase I – Non-Associative and Pavlovian (*classical*) Conditioning:** For the first few weeks of the course, you will work with Sniffy, a virtual rat, by using your own computer and during your own time. **You must submit your Sniffy assignments at the time that they are due. Additionally, if you fail to do these Sniffy projects and their assignments, you will find it very difficult to pass the Sniffy exam (*see calendar for the date of this exam*). Finally, if you fail to submit 5 or more of these Sniffy assignments, you will not be allowed to take the Sniffy exam. Failing to do the minimum Sniffy assignments and thus becoming ineligible to take the Sniffy exam will significantly hurt your course grade (*see below for the percentages of these events*).**
- If you do not have access to a computer, talk to your Learning & Behavior lab TA. By coordinating with a lab TA, you might be able to use to the computer in the lab (BSB, A323) for your Sniffy activities.
- **Phase II – Operant Conditioning:** After Phase I is complete and for the rest of the semester, operant conditioning experiments are conducted in Room A323 of the Baylor Sciences Building (BSB) and during each partner's specific lab time. In these experiments, you and your partner will share and train a live

rat using positive reinforcement. Most of this lab manual is concerned with this phase of the lab, with the exception of those pages that specifically discuss Phase I and the Sniffy exercises.

Important General Notes on the Pavlovian Conditioning Phase (*Phase I*) of this Lab:

Sniffy Project Submissions:

- Your Sniffy project submissions and your answers to any tasks and/or questions (*if applicable*) must be turned in to the graduate students, on Canvas, or email (*only if necessary*).
- In the “Task(s)” sections for the Projects that follow, you are often asked to interpret the results of your Sniffy exercises. To get maximum credit for these projects, you must clearly demonstrate that you understand your results. **In science, a few thoughtful and well-written sentences, organized into competent and coherent paragraphs, suffices.**
- Each of your submissions MUST identify the project number and the specific exercises conducted for that project. For example: label your submission “*Project 5: Experiments 7 & 8*” and include the graphs for these exercises, along with your interpretation of the data, as well as your answers to any questions that were posed for the project, if applicable.

Important General Notes on the Operant Conditioning Phase (*Phase II*) of this Lab:

General Comments:

- During Phase I (Sniffy), you will be given a lab partner and the two of you will work together during Phase II (Operant Conditioning). Obviously, your partnerships must be established before Phase II begins with Experiment 1 (Handling).
- You will attend the lab twice a week for one-hour lab sessions.
- Your lab TA will upload the phase II experimental data to Canvas, which you will then access and use to construct your graphs.
- You and your partner’s lab sessions will alternate (*MW and TR*) and your respective lab sections must occur at relatively the same time.
- We may have to schedule additional sessions on Fridays, but **only if absolutely required** (*e.g., weather cancellation days, equipment malfunctions, scheduling issues, poor performance by the rat, experimental objectives not met, etc*).
- You must take note of your lab instructor’s name and email (*write it down in your lab notes*) so you can contact them with attendance issues, or with any questions and/or problems that you might have.
- You must also note your lab partner’s email address so that you can communicate with them throughout this lab course.
- The author of this manual and the person who created these computer protocols/experiments is Dr. Hugh H. Riley, his cell phone is 430-236-3916 and his email is Hugh_Riley@Baylor.edu.

THE LEARNING & BEHAVIOR LAB POLICY ON ATTENDANCE, ABSENCES AND PUNCTUALITY

Attendance:

- Laboratory work, especially with live animals, requires strict discipline, and in this course, you **cannot miss or be tardy to any of your assigned sessions**. The extract below, taken from Baylor’s Student Policies and Procedures website, clearly identifies that faculty can establish such additional attendance criteria:

University policy requires that the student must attend at least 75% of all class meetings. **Faculty members may establish additional attendance requirements as outlined in course syllabi.** Excessive lateness will be dealt with by the individual instructor. All course requirements may not necessarily be fulfilled by meeting the 75% minimum. The student bears the responsibility for the effect which absences may have upon class participation, announced and unannounced examinations, written assignments, reports, papers, and other means of evaluating performance in a course.

The Learning & Behavior Lab Policy on Absences and Punctuality:

Note in the official statement above regarding absences and tardiness, that faculty can impose additional absence and tardy policies. In this lab, if you must be absent from the lab for whatever reason, you **must** contact your lab instructor to work out a specific time to make up that lab session. Additionally, **you must contact your lab instructor WELL BEFORE your absence or tardy event so that your rat will receive water for that day.** If you are absent for a non-emergency reason without notifying anyone, and if your rat is subsequently harmed by dehydration, **you will fail this course.** You will also not receive data for that portion of the experiment and you will lose points on your assignment. You may also be asked to reimburse the Department of Psychology & Neuroscience for the cost of your rat. Further, these notifications **MUST** be made directly to the TA and in person (*preferably*) or by email. Telling a friend or leaving a voicemail on the Department of Psychology office phone do NOT constitute *PRIOR* notification. Additionally, you should understand that absences make it impossible to submit the assignments on time and it also inconveniences your classmates and lab instructor. Moreover, your lab partner will not have data for the days that you are absent and that is not fair to them. For all of these reasons, and as previously stated, **you simply cannot be absent from any of your scheduled lab sessions during Phase II**, unless the absence occurs for a very important, unavoidable, and serious reason. You can also call Dr. Riley (*cell phone: 430-236-3916 email: Hugh_Riley@Baylor.edu*). Failure to keep your lab instructor informed will severely impact your course grade (*see below for more specific information*).

The Following Absences are Excusable:

- The student had a death in their immediate family (siblings, parents, grandparents)
- The student was too ill to come to class or is concerned they may have COVID (they must have a physician's letter/ COVID test confirming this)
- The student was unable to take the exam because of their attendance at an official Baylor activity

Penalty for Absences and/or Lack of Punctuality:

- There are eight (8) operant conditioning chambers in the laboratory (*BSB, A.323*) and all of these stations start at the exact same time during each lab session. For that reason, any student who is late holds up the entire class, and we cannot stop the trial halfway through the experiment to accommodate anyone who arrives late. **Therefore, at three (3) minutes after the lab session is scheduled to begin, your lab teaching assistant (TA) will lock the door. At that point, anyone who arrives after the door has been locked will not be allowed to enter the room nor will they be allowed to participate in any of that session's activities, including submitting their assignments and taking the quiz.** Consistent tardiness WILL NOT BE TOLERATED (see below).

- If your absence does not fall under any of the “excusable” circumstances but you give your instructor a **PRIOR notification, 5 points will be removed** from your next operant conditioning assignment.
- If you do **not give me a PRIOR notification, 20 points will be removed** from your next operant conditioning assignment.

What You Should Do If You Must Be Absent or Will Be Unavoidably Late:

- Make sure you contact your instructor and explain why you will be absent/tardy. You can also call Dr. Riley (*cell phone: 430-236-3916 email: Hugh_Riley@Baylor.edu*). To avoid the penalty listed above for a legitimate absence or tardy episode, this notification **MUST OCCUR WELL BEFORE** your lab session is scheduled to begin.
- If you are too ill to attend a lab session, you must submit a doctor’s letter at the next lab session (or a picture of your COVID test).
- You must also arrange for someone (other than your TA) to run the experiment for that day so that your partner will have the data for that session.
- You should also ensure that your rat receives water after the session. This person can be your lab partner or someone else in the course. Do not expect your TA to be able to help you as they have other duties.

Your rat's life is dependent on you and your lab partner's attendance in your respective lab sessions. Therefore, your attendance in these lab sessions is mandatory.

EVALUATION (GRADING CRITERIA)

- **Overview:** As previously stated, NSC/PSY 3120 is a one credit hour laboratory course that is graded independently from the learning & behavior lecture class (*NSC/PSY3320*). Your course letter grade in this lab is based on the well-being of your rat, your attendance, your quizzes and exams, and on the quality and timeliness of your submitted assignments.
- **Quizzes:** Before each operant conditioning experiment begins (or during the experiment itself), you will be quizzed over the assigned readings in your Lab Manual. These quizzes will be taken through CANVAS during class, so a personal computer is required. If you do not have a laptop please inform your TA. The quizzes contain multiple choice and true/false questions, and they are designed to test your understanding of the learning and behavior phenomena that you will investigate during the experiment. To do well on these quizzes, you must be thoroughly familiar with the experiment design and its purpose **BEFORE** the experiment begins. As shown below, these quizzes contribute 20% to your lab course grade.
- **Operant Conditioning Assignments:**
 - Periodically throughout the semester (*see the calendar for due dates*), you will analyze yours and your partner’s data on the operant conditioning experiments and submit those assessments to Canvas. As shown below, these assignments are worth 25% of your lab course grade. See below for more information on these assignments.
 - As just stated, all of your phase II assignments **MUST** be turned in by the due date (*late submissions will not be accepted*). You cannot give them to the lecture course (*NSC/PSY3320*) professor(s), nor can you drop them off in the Department of Psychology office.
 - When completing your assignments, if your data is partially incomplete for some reason, **DO NOT** substitute some other rat’s behavior for your rat’s behavior. In science, we don’t make the

graphs look pretty by modifying or manipulating the data – we simply **REPORT AND EXPLAIN** our results. In other words, if your operant conditioning chamber malfunctioned for some reason and there's no data for that particular day, then neither your data nor your graph should show any activity on that particular day, and you **MUST** explain these results in your analyses.

- There is one exception to this policy. If you don't have any data whatsoever for some reason, then you must explain why and you must also get some sample data from your TA to graphically represent. In such cases then, you must submit two sets of data, two graphs, and two analyses.
- If your analyses of your data do not match and explain your results on an assignment, you **will not receive any credit** for that particular assignment. In other words, your data, their graphs, and your analysis must reflect and relate to each other and your analyses must be based on the data and its graphical illustration.
- **Attendance & Tardiness:** See the discussion above.
- **Data Recording in Your Lab Manual:** You must keep a record of your animal's weight and water consumption during every lab session both in the log kept in the lab room (A.232) and your personal log. The last page of your Lab Manual contains a template for this log, which you should print out and bring with you to every lab session.
- **Exams:** There are two major exams in this lab: a comprehensive exam on Phase I (*non-associative and classical conditioning using Sniffy*), and a final exam over Phase II (*operant conditioning with a live animal*). As shown below, collectively, these exams contribute 20% of your lab course grade.
- **Rat Olympic Preliminary Write-Up and Final Report:** You and your partner will design an activity for your rat and then train it to perform this behavior using positive reinforcement (*see the calendar for the dates of this activity*). Collectively, these Rat Olympic reports contribute 20% of your lab course grade.
- **Bonus Points:** Completing both the course evaluation at the end of manual and Baylor online course evaluations you will receive 10 extra points on your final grade.

Summary of the Grading Criteria:

Assignments, Quizzes and Exams	Percentage	Points
Sniffy project assignments	10%	100
Sniffy™ The Virtual Rat Exam	10%	100
Phase II Quizzes	25%	250
Operant Conditioning Assignments	25%	250
Rat Olympic Preliminary Write-Up	10%	100
Rat Olympic Final Report	10%	100
Laboratory Final Exam	10%	100
Total	100%	1000

Phase I: Non-Associative Learning & Classical Conditioning (*Sniffy™, the Virtual Rat*)

The Sniffy Virtual Rat program is used extensively and often exclusively in many Universities for investigating learning & behavior phenomena in an accompanying lab for a learning and behavior course. Therefore, you are very fortunate to also have the opportunity in this lab to work with a live animal to investigate operant conditioning. However, before we can begin those operant conditioning experiments, you should have a thorough understanding of non-associative (*single stimulus*) learning and classical conditioning, and these Sniffy projects should help you better understand these learning phenomena. **To accomplish these Sniffy projects, you must load the Sniffy program onto your own personal computer and then complete these projects on your own (*The SniffyPRO app must be purchased and used*). This must be accomplished prior to the first week of class, or you will become behind.** If you don't have access to a personal computer for these Sniffy projects, we can allow you access to the Baylor computer in the Learning and Behavior Laboratory. In the event that you need this service, please contact your instructor, Dr. Riley, or one of the other instructors so that we can schedule a time for you accordingly.

Your lab TA might choose to meet with your section to discuss the concepts covered by the Sniffy Virtual program (*each TA makes this decision for their lab sections*). If so, your TA will answer any questions you may have about these Sniffy activities, and they might also demonstrate the experiment to you. If your TA chooses this activity, you should use this opportunity to be sure you understand how to run the program, print your results, and fully understand the concepts in these activities as they are tested in the Sniffy exam.

General comments about the Sniffy program:

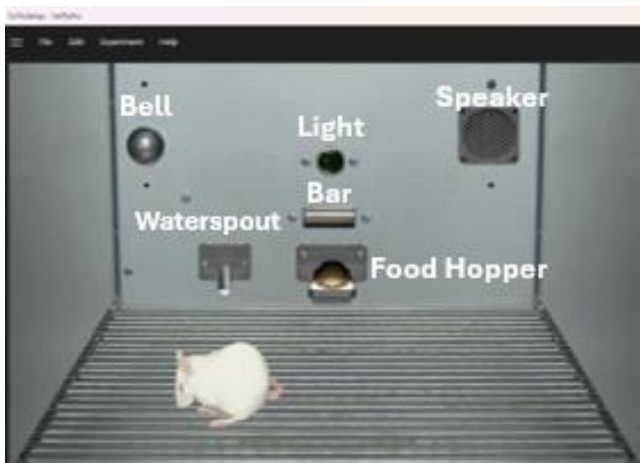
- If you experience any technical difficulties with the Sniffy program, you can contact the Baylor technology help desk or Moody Library front desk for help.
- In most of the exercises in this Sniffy manual, Examples are provided of what "*should happen*" to Sniffy's behavior and what your graphs *might* resemble when the exercise is completed. However, your graphs will not (*and should not*) exactly match these Examples because every Sniffy is different.
- **Do not skim this Sniffy manual text** – failing to read the specific instructions for an exercise can cause real difficulties in understanding the Sniffy graphical displays and properly interpreting your results. Besides, the Sniffy manual does an excellent job explaining many learning principles.
- In Sniffy graphs, windows with a light blue background represent theoretical variables (*i.e., inferred mental states*), while those windows with a white background display actual behavioral data (*e.g., the Movement Ratio and Cumulative Record windows*).
- In these classical conditioning experiments, a common misunderstanding involves the distinction between stages and trials in the experimental design. An experiment may involve one or more stages, and another stage, which are represented by numbers (*e.g., 1, 2, 3, etc*) may consist of one or more trials, which are represented by a letter (*e.g., A, B, C, etc*).
- In the Cumulative Record window for any Operant conditioning experiments that you might elect to perform, each upward movement of the dots toward the top of the graph represents one response, while each diagonal mark of the dots represents the delivery of a reinforcer.
- The sound can be turned off and on. To do this, simply click on File/Preferences and then check the box next to "Soundproof Cage."

- The amount of time required to perform these projects depends on the speed of your computer and the speed setting of the Sniffy animation (*i.e., you can and should accelerate time in many of the exercises*). Most of the exercises actually take very little time to run.
- Remember to ARCHIVE all data files because some of these projects build upon previous ones.
- Please take careful notes on any difficulties that you encounter with these projects or this lab manual and bring them to Dr. Riley's or your TA's attention.

What You Need to Know About Sniffy

Although using a real rat in an in-person lab format is nice, using the virtual Sniffy program has some really great benefits. One of the main benefits is cost. Using a virtual computer program saves on the expenses of keeping a rat alive and the equipment necessary for an in-person experience. Using Sniffy also allows for students to observe some learning experiences that a real rat lab could not provide. For example, a student is able to see the graphs associated with the rat's interactions move in real-time. A student is able to see the rat's fear of a stimulus grow or diminish while it is occurring in the Sniffy program while a real rat would not provide that kind of detail. The Sniffy program is designed to be as realistic as possible and even allows for the rat to become tired if an experiment is taking too long. The principles and ideas learned from using Sniffy, can and should be applicable to real-world and human scenarios. There will be step by step instructions for each experiment under the projects labeled below.

Operant Chamber: The operant chamber that is used in this program closely resembles the basics of what most scientists use. The rat walks on bars so that feces and urination can easily fall below to the dustpan and not become unsanitary for the rat. There is a waterspout and food hopper on the back wall for water and food to be dispensed. There is also a light, bell, and speaker on the back wall to be used during different conditioning sessions. A labeled picture is below.



Classical Conditioning Background

Classical conditioning is caused by two different stimuli occurring in a regular and reliable sequence allowing one stimulus to predict the other stimulus. Another name for this is Pavlovian conditioning, because Ivan Pavlov was one of the first people to experiment with this type of learning. In classical conditioning the stimulus that comes first is the conditioned stimulus (CS), which is followed by the unconditioned stimulus (US). The US is the stimulus that elicits a noticeable and measurable response called the unconditioned response (UR). Most people do not measure the CS because the response is often too small to measure; therefore, the US is what is measured the majority of the time. Once the CS and US are paired enough times the CS will begin to elicit a new response called the conditioned response (CR). The best-known example of this type of learning is Pavlov's dogs. Pavlov used food as the US which caused drooling and salivation in the hungry dogs (UR). He repeatedly paired the food with a tone (CS) until the tone (CS) caused the hungry dogs to drool and salivate even when there was no food present (CR). It will be

HELPFUL to know that in the Sniffy program, the movement ratio window depicts a change in the rat's behavior while the CS Response Strength mind window shows the change in the rat's learning parameters (also known as the rat's psychological state influencing its behavior). For example, if the movement ratio is increasing, it shows that the CS is causing freezing behaviors. As the movement ratio increases, the CS response strength will begin to illustrate that the CS's ability to cause fear is increasing as well.

NOTE: *It is recommended that you read the instructed "Task(s)" section of each project prior to beginning the assignment, this will allow you gather the required screenshots and data as you go if preferred. Remember to title ALL of your files, including screenshots, to keep your information organized and accessible to you.*

Non-Associative Learning

Sniffy Projects 1-3 Instructions

In this assignment you will create a single document for all three projects you have completed with the Sniffy software. Please be sure to read all project instructions carefully and do not skip over sections, otherwise, your report may be incomplete.

Project 1 looks at Sniffy's baseline behaviors, **Project 2** teaches about habituation and sensitization, and **Project 3** assesses the process of acquisition, extinction, and spontaneous recovery.

- For Project 1 you will:
 - Submit the Excel file of your observational record of Sniffy's baseline behavior.
 - View the video lecture on Canvas on the "Pyramid of Learning".
 - View the video lecture on the importance of baseline.
- For Project 2 you will:
 - Submit the Movement Ratio and the CS Response Strength windows from each of the 3 shock conditions (low, medium, and high) for exercise 19.
 - Interpret the Movement Ratio and CS Response Strength results.
 - Your interpretation should be separate for each graph and very thorough if you want to receive full credit.
- For Project 3 you will:
 - Submit the document for Exercise 3 (I will see Exercise 1 and 2 as well, since exercise 3 is based on having run Exercises 1 and 2), exercises 4 and 5 and submit the Movement Ratio & CS Response Strength windows for Exercises 3, 4, & 5.
 - Interpret these results.
 - View the video lecture on the importance of classical conditioning.

Project 1: Establishing a Baseline of Behavior.

One common way to view whether learning has occurred is to compare what an animal does before it learns something (*i.e., before conditioning occurs*) with what it does after it has formed an association between two things (*e.g., either two stimuli in the case of classical conditioning, or a response and its consequence in the case of operant conditioning*). Therefore, before any conditioning occurs, it is very important to know an animal's behavioral baseline (*i.e., what it prefers to do normally*). In this project, you will establish this behavioral baseline by simply observing Sniffy's natural (*unconditioned*) behaviors. For

this project, your goal is simply to establish a baseline of their pre-conditioning (*preferred*) behaviors. To accomplish this, watch Sniffy for ten minutes, and then using Excel, create a list of 4 to 5 behaviors that you see Sniffy do, and then simply record the number of occurrences for each of these behaviors. This is just Sniffy in the chamber, so you do not need to set up an experiment.

Background: Similar to humans, rats behave in predictable patterns when under certain “emotional states.” For example, if a rat is scared it will often sit in a corner and be very still (this is called freezing behavior). If a rat is in a new environment, it will explore the environment by sniffing around, rear/stand up, and walk around. When a rat is calm it will often perform grooming behaviors, such as wipe its nose with forepaws, scratch, and lick/groom itself.

Example:

Behavior	Occurrences
Sniff	24
Stand up	16
Scratch nose	26
Pick at hind legs	10

Task(s):

- Submit the **Excel file of your observational record** of Sniffy’s baseline behavior.
- **View the video lecture** on Canvas on the “Pyramid of Learning”.
- **View the video lecture** on the importance of a baseline.

Project 2: Habituation and Sensitization

Classical and operant conditioning reflect associative learning, but as will be discussed in lecture, biology has determined that animals must be sensitive to a single stimulus (*non-associative*) because it promotes their survival. Examples of single stimulus learning include reflexes, fixed or modal action patterns, habituation, and sensitization. In this experiment you will learn the differences between habituation, sensitization and background conditioning.

Note: When running Project 2, you SHOULD use the “*Isolate Sniffy*” option to accelerate time. If you don't do this, each of the conditions will take about 8 hours to run, which is too time consuming. However, if you do isolate Sniffy, things happen quickly, which means that you might run to the end of the time limits of the program (*Sniffy projects are time limited, which simply means that at the end of that time, Sniffy goes to sleep and you can't wake them up*). If this occurs, you cannot do anything with Sniffy once they start napping, so your only option is to close that Sniffy file, start a new one, and this time don't leave Sniffy in accelerated time for too long.

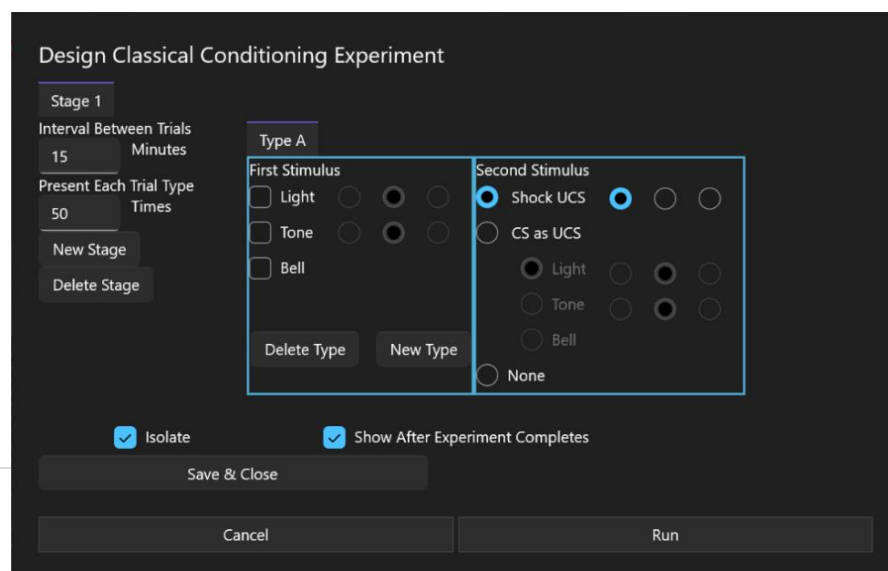
Terms: When referring to Sniffy's UR, that's their **Unconditional Response** - what they do automatically. Think about what your UR to being shocked would be - Sniffy's is very much the same. US is an **Unconditional Stimulus** - one to which you react automatically - the US here is the shock. CS is the

Conditional Stimulus - one to which you learn to react because of training. Here the US (shock) has no stimulus right before it, so Sniffy will learn to fear the environment (the cage). With these terms under control, the Manual will make sense.

Background: If a rat is shocked without a prior CS a couple different outcomes can occur based on the type of shock received. If the shock is low intensity, the more often the rat is shocked, the less the rat will respond to the shock. Eventually the rat will stop responding to the shock and continue doing whatever it is doing- this phenomenon is called **habituation**. If the shock is medium intensity, the rat's behavior will be disrupted for around two minutes every time it is shocked and then go back to normal. The rat's reaction will not vary too much. However, if a high intensity shock is given, the rat will continuously have larger and more prolonged disruption to their behavior. At first the rat will be disrupted for about two minutes but will steadily increase to around 4 minutes. This type of behavior demonstrates **sensitization** to a CS. If a medium or high intensity shock is given without any prior CS, they will slowly develop **background conditioning**; meaning that the rat will form CR's to their context. A rat could develop a CR to the cage wall or floor so that after a long time, they would be in continuous fear of a shock due to always being around the environment they were shocked in previously. This can eventually cause "fear paralysis" as the rat will spend the majority of their time frozen in fear of a shock.

Running the Experiment: In this experiment, you will have 3 different stages: 50 low intensity US, 50 medium intensity US, and 50 high intensity US. To begin you will need to follow the steps below:

1. **Open a new Sniffy file and immediately give it a new name. Save it in a Sniffy file folder on your computer.** We suggest that you save a separate one for low, medium and high intensity US's.
2. **Under the experiment tab, click the 'Design Classical Conditioning Experiment.'**
 - a. **Set the "Interval Between Trials" to 15 minutes**
 - b. **Set the 'Present Each Trial Type' to 50 times**
 - c. **Make sure nothing is selected in the 'First Stimulus' box and select either low, medium, or high intensity in the "Second Stimulus" box (you will end up doing all three).**
 - d. **Your set up should look like this:**



3. Make sure the 'Isolate' button is clicked and the 'Show After Experiment Completes' is clicked as well, then push run and SAVE THE FILE.
4. Once completed, copy your results for the Sensitivity and Fear graph, Movement ratio graph, and the CS response strength graph and put them on your assignment document. Your graphs may appear little to no graphical information for the movement ratio and sensitivity & fear graphs when set to the low intensity setting – that is okay! However, if these are blank for medium and/or high intensity, this is likely a technical error.

How to Copy Graphs:

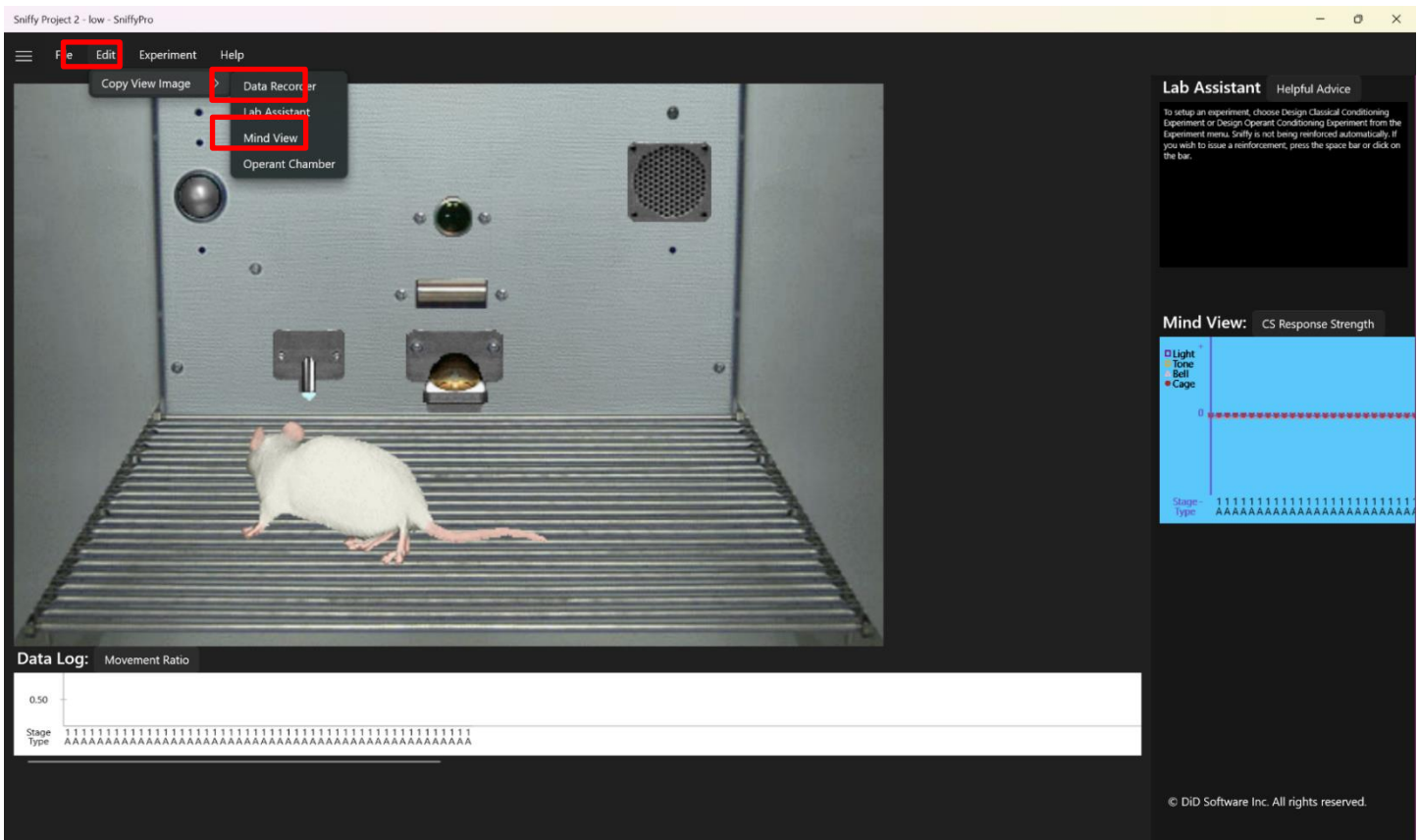
1. Make sure the Mind View and Data Log windows are set to the correct graph you want.

2. Under the **Edit** tab, move your mouse over "Copy View Image."

3. Choose which window you need to copy (Data Recorder or Mind View).

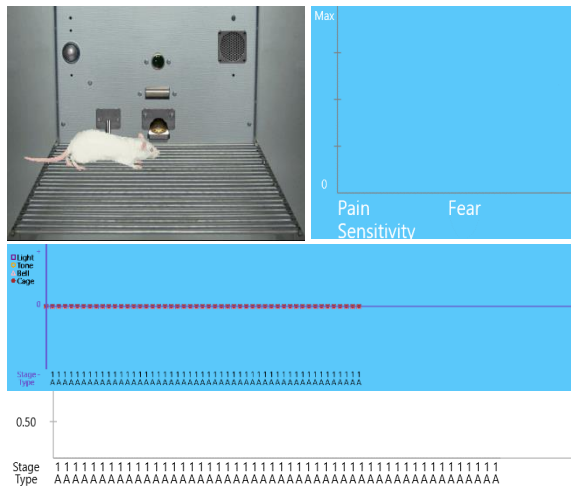
- For this assignment you will also need the Pain and Fear Sensitivity Graphs which are located under the mind view window (meaning you will copy the Mind View image twice)
- If you want to include a picture of sniffy (not required) then you can copy the operant chamber image.

1. The photo of your graph is now copied to your clipboard where you are able to paste it to your assignment document.

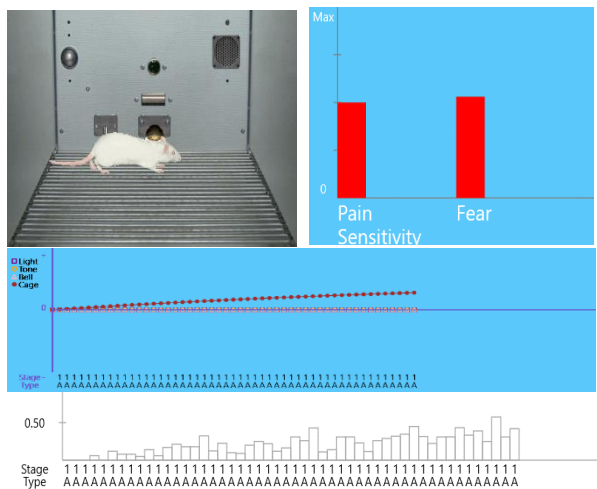


Example Graphs:

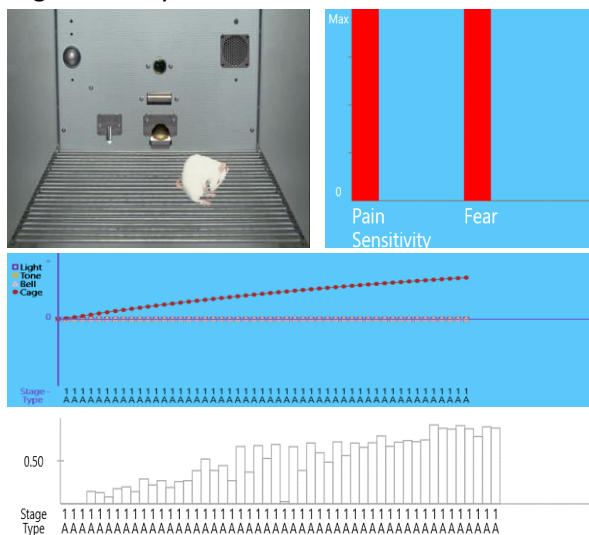
Low Intensity:



Medium Intensity:



High Intensity:



Task(s):

- Submit **the Movement Ratio, Sensitivity & Fear, and the CS Response Strength windows** from each of the 3 shock conditions for Project 2.
- **Interpret** the Movement Ratio and CS Response strength results and submit.
- **Submit** your interpretation of your results. Your interpretation should be separate for each graph and very thorough if you want to receive full credit.
- *NOTE: These assignments should be clean and organized, with each graph having a clear title, axis, etc. Failure to do so will result in loss of points.*

Classical Conditioning

Project 3: Acquisition, Extinction, Spontaneous Recovery (Exercises 1, 2, 3), and Varying the Strength of the CS and US (Exercises 4, 5)

This project introduces Classical Conditioning. In our previous project, there wasn't a specific cue (*other than time*) that told Sniffy when the shock would be delivered. Here, you will associate a specific cue with a shock and you might see a difference in their behavior. Complete Sniffy exercises 1, 2, 3, which deal with acquisition, extinction, and spontaneous recovery. After that, complete Sniffy exercises 4 and 5, which deal with varying the strength of the CS (exercise 4) and varying the strength of the US (exercise 5). For exercises 1, 2, and 3, basically you will run exercise 1, save it, and then use that file to run and save Exercise 2, which you will then use to run Exercise 3, which you will also save. In these experiments you will learn about the acquisition, extinction, and spontaneous recovery of learning in a rat model.

Terms: When referring to Sniffy's UR, that's their **Unconditional Response** - what they do automatically. Think about what your UR to being shocked would be - Sniffy's is very much the same. US is an **Unconditional Stimulus** - one to which you react automatically - the US here is the shock. CS is the **Conditional Stimulus** - one to which you learn to react because of training. Here the US (shock) has no stimulus right before it, so Sniffy will learn to fear the environment (the cage). With these terms under control, the Manual will make sense.

Background: Acquisition learning is the process of continually pairing the CS with the US in an experiment. The repeated pairing of these two stimuli results in the CS acquiring the ability to elicit a CR by itself. Once a CR has been acquired, it can be extinguished by continuously pairing the CS alone with no US; this is called **extinction**. Even after the animal's learned response has been eliminated, it can return. If a rat is removed from the learning environment for a few days and then placed back in the same setting, it is highly likely that the CR will occur during that learning session; this is called spontaneous recovery. In fact, during the second extinction session, the CR will be stronger than the end of the first session. The movement ratio graph will show the rat's behavior while the CS response rate graph depicts the "psychological state" that influences the behavioral changes in the rat. For example, an increase in the movement ratio means that the CS is developing the ability to create fear behaviors; as this graph increases, the CS response should show that the CS's ability to create fear behaviors is increasing as well. However, if performing an extinction

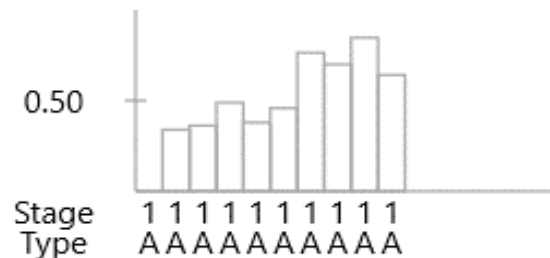
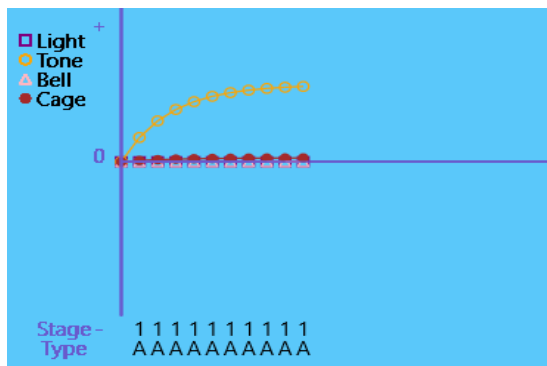
experiment, the movement ratio would illustrate that the repeated pairing of a CS with no US causes a gradual stop of the fear behaviors such as freezing. The CS response strength would enhance this by showing the CS is losing its ability to create a fear response within the mouse.

Running Exercise 1: Acquisition is created by consistently having the CS precede the US in multiple trials.

1. **Open a new Sniffy file and immediately give it a new name. Save it in a Sniffy file folder on your computer.**
2. **Under the experiment tab, click the 'Design Classical Conditioning Experiment.'**
 - a. **Set the "Interval Between Trials" to 5 minutes.**
 - b. **Set the 'Present Each Trial Type' to 10 times.**
 - c. **In the 'First Stimulus' box select medium intensity tone.**
 - d. **In the "Second Stimulus" box select the medium intensity shock.**
3. **Make sure the 'Isolate' button is clicked and the 'Show After Experiment Completes' is clicked as well, then push run and SAVE THE FILE. You do not need to copy your graphs yet.**

It is recommended that you save your data at this point (see "Create three data export files" below). Otherwise, you will need to reopen your file to export the data later.

Example Graphs:

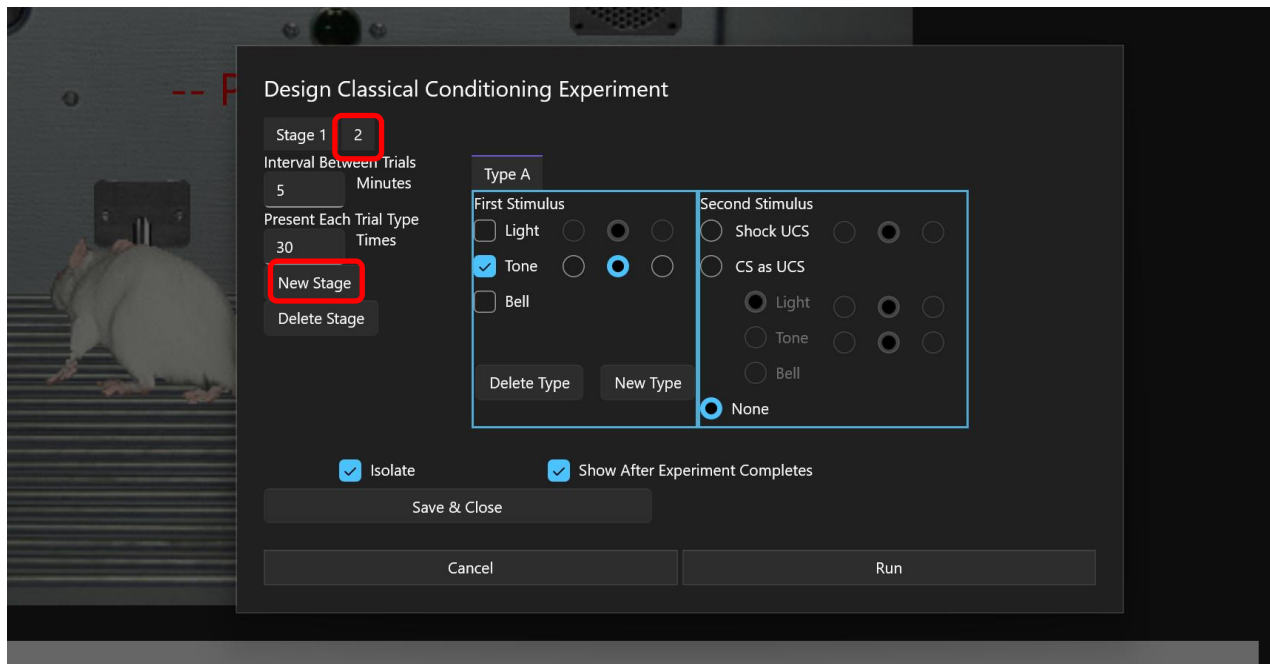


Running Exercise 2: This experiment builds from experiment one. It is good to note that you will always need to run more extinction trials than acquisition trials because the CR diminishes slower than it is learned.

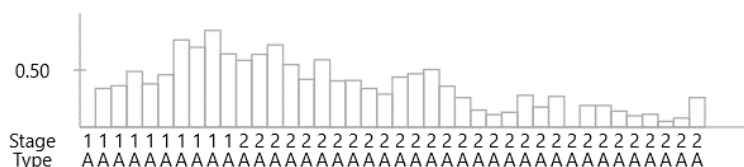
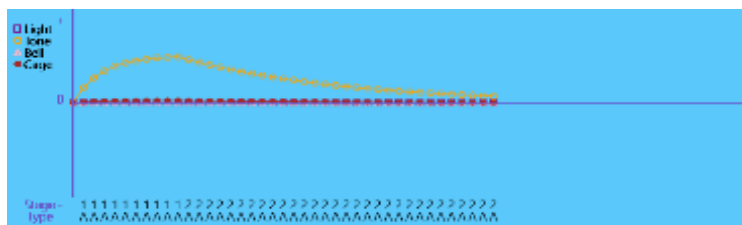
1. **Open the file used for experiment one and save it under a new name (like EX2-Ext).** By saving this file under a new name, you can still have the experiment one file while using its data for this experiment.

2. **Under the experiment tab, click the ‘Design Classical Conditioning Experiment.’** This will appear dimmed because it was already used for experiment one. You will need to click **‘New Stage’** in order to change the settings and continue.
 - a. **Set the “Interval Between Trials’ to 5 minutes.**
 - b. **Set the ‘Present Each Trial Type’ to 30 times.**
 - c. **In the ‘First Stimulus’ box select medium intensity tone.**
 - d. **In the “Second Stimulus’ box make sure ‘None’ is selected.**
3. **Make sure the ‘Isolate’ button is clicked and the ‘Show After Experiment Completes’ is clicked as well, then push run and SAVE THE FILE.** You do not need to copy your graphs yet.

Example Window:



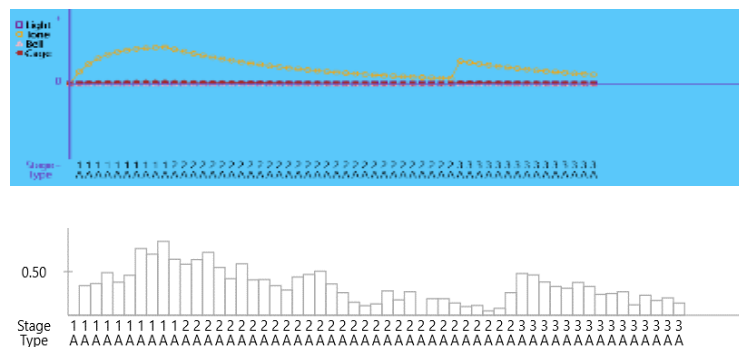
Example Graphs:



Running Exercise 3: The movement ratio graph depicts the function of trials while the CS Response Strength will represent the tone's ability to produce a fear response in the rat.

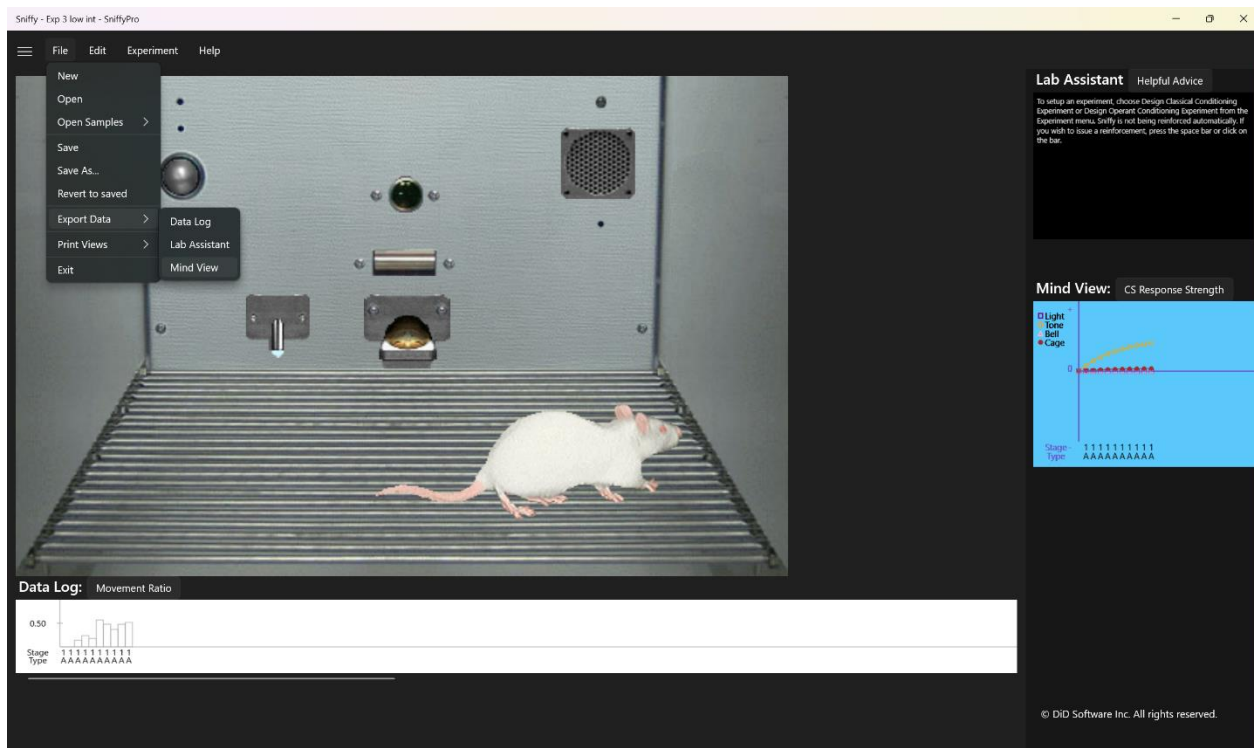
1. **Open the file used for experiment two and save it under a new name (like EX3-SR).** By saving this file under a new name, you can still have the experiment one file while using its data for this experiment.
2. **Under the 'Experiment' menu, click "Remove Sniffy for Time Out.'** This will imitate removing the rat for a 24-hour period of rest. **To put the rat back into the operant chamber, click the 'OK' button in the dialogue box.**
3. **Under the experiment tab, click the 'Design Classical Conditioning Experiment.'** This will appear dimmed because it was already used for experiment one and two. **You will need to click 'New Stage' in order to change the settings and continue to Stage 3.**
 - a. **Set the "Interval Between Trials' to 5 minutes.**
 - b. **Set the 'Present Each Trial Type' to 15 times.**
 - c. **In the 'First Stimulus' box select medium intensity tone.**
 - d. **In the "Second Stimulus' box make sure 'None' is selected.**
4. **Make sure the 'Isolate' button is clicked and the 'Show After Experiment Completes' is clicked as well, then push run and SAVE THE FILE.**
5. **Once completed, copy your results for the Movement ratio graph and the CS response strength graph. You will only turn in this experiment's graphs because they contain all the graphs from experiments 1-3.**

Example Graphs:



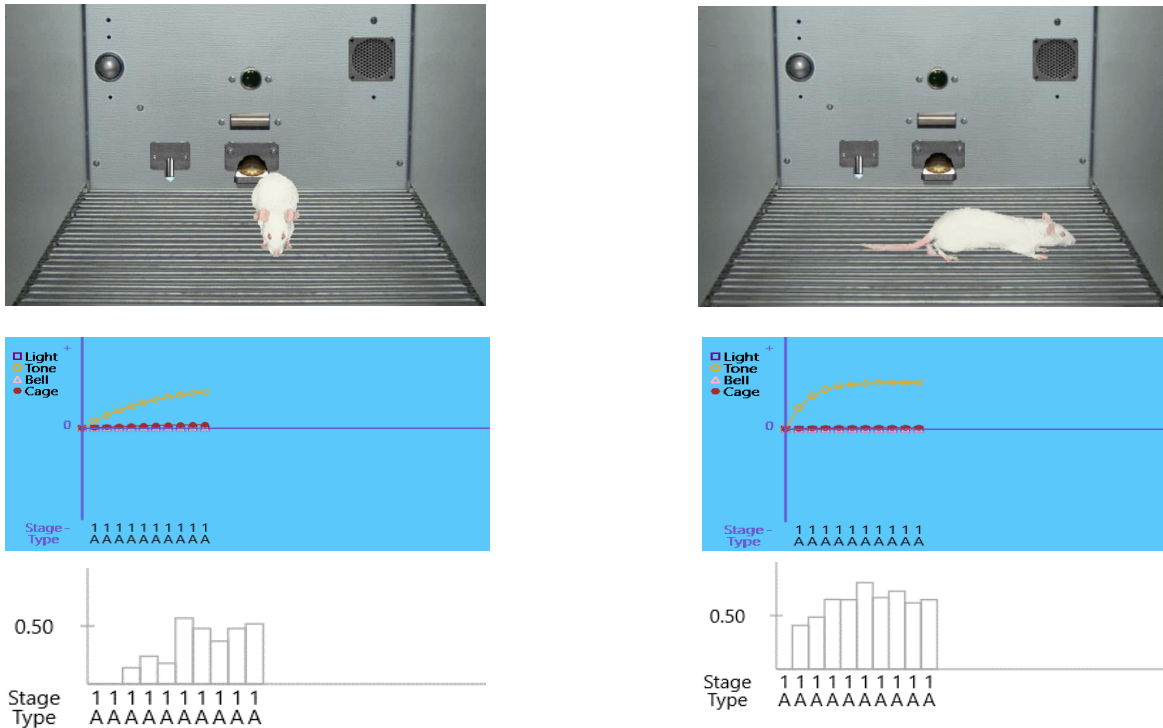
Running Exercise 4: It is important to note that the strength of the CS strongly determines the time required to acquire classical conditioning. The higher the intensity of a CS, the faster the acquisition will take. This experiment will compare the medium intensity tone in experiment one with a low and high intensity tone.

1. **Open a new Sniffy file and immediately give it a new name. Save it in a Sniffy file folder on your computer (For example, ex4-lowtoneCS).**
2. **Under the experiment tab, click the ‘Design Classical Conditioning Experiment.’**
 - a. **Set the “Interval Between Trials’ to 5 minutes.**
 - b. **Set the ‘Present Each Trial Type’ to 10 times.**
 - c. **In the ‘First Stimulus’ box select low or high intensity tone (You will do both).**
 - d. **In the “Second Stimulus’ box select the medium intensity shock.**
3. **Make sure the ‘Isolate’ button is clicked and the ‘Show After Experiment Completes’ is clicked as well, then push run and SAVE THE FILE.**
4. **Copy your movement ratio and CS response strength graph.**
5. **Export your data.** Go to File -> Export Data -> Mind View. Make sure the CS Response Strength is open in the Mind View window. **Save your data** under a similar name to the exercise. You will need this file to complete your assignment.



6. Once the low intensity has been done, repeat these steps to do the high intensity tone. Next, you will need to make your own graph using excel.

Example Graphs: (The left image is low intensity, and the right is high intensity).



Hint: The leveling out of the CS Response Strength curve demonstrates that the intensity of the CS only affects the speed at which it is learned, not the level of learning reached.

Create three data export files:

1. Open the correct Acquisition Sniffy File (you will need exercise 1, exercise 4 low and exercise 4 high).
 - a. Once the file is open, make sure the 'Mind View' is open and display the CS Response Strength graph.
 - b. With the CS Response Strength window selected: Under 'File', click the 'data export' tab and choose the 'Mind view.' Save the file in a folder with the other Sniffy files somewhere that is easily accessible. Do this for all three: low, medium (exercise 1), and high intensity, and then close Sniffy.
2. Open excel and save the file in the same area as your other Sniffy files.

3. Enter row and column headings to create a group of cells that resembles the following:

4. Open your three Sniffy data export files. To open each file in your spreadsheet program:

Trials	Weak CS	Medium CS	Strong CS
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

- Choose the Open command from that program's file menu. In the dialogue box that appears, **go to the location on your hard drive where you saved the Sniffy Pro data export files**. If the Sniffy Pro data export files are not visible, choose the 'Show All Files' option in the file-opening dialogue box. If no 'Show All Files' option is available or if the program does not open the file successfully when you select it, check to see whether the program has a special data import command. If necessary, read your spreadsheet program's manual or search its online Help files to determine how to import a tab-delimited text file.

5. Repeat this process for the other two data export files. Each of the three data export files will resemble the next figure when opened in your spreadsheet program:

	A	B	C	D	E	F	G	H	I	J	K	L
1	Stage	Trial	S/R	Light	Tone	Bell	Background	BP during	BP pre-CS	Freeze During CS	Active During CS	Movement Ratio
2	1	A	0	0	0.3	0	0.015	0	0	0	300	0
3	1	A	0	0	0.5055	0	0.0253	0	0	84	217	0.2791
4	1	A	0	0	0.6463	0	0.0323	0	0	84	217	0.2791
5	1	A	0	0	0.7427	0	0.0371	0	0	165	135	0.55
6	1	A	0	0	0.8087	0	0.0404	0	0	132	169	0.4385
7	1	A	0	0	0.854	0	0.0427	0	0	150	151	0.4983
8	1	A	0	0	0.885	0	0.0442	0	0	170	131	0.5648
9	1	A	0	0	0.9062	0	0.0453	0	0	249	52	0.8272
10	1	A	0	0	0.9208	0	0.046	0	0	204	97	0.6777
11	1	A	0	0	0.9307	0	0.0465	0	0	220	81	0.7309

Hint: The data are presented in vertical columns headed by the letters A through L and in horizontal rows headed by the numbers 1 through 11. There are 12 columns of data. In Row 1 of each column is a label that tells you what kind of data appears below. The column labeled Stage lists the stage of the experiment for each trial. Because this experiment had only one stage, the numeral 1 is entered for every trial. The column headed Trial gives the trial type for each trial. Because this experiment had only one trial type, the letter A appears for every trial. The column called S/R gives the suppression ratio for each trial. The columns called Light, Tone, Bell, and Background give the CS response strength values, respectively, for the light CS, tone CS, bell CS, and background (cage) CS for each trial. The column labeled BP During CS tells how many times Sniffy pressed the bar while the CS was on during each trial. The column labeled BP Pre-CS tells how many times Sniffy pressed the bar during the 30-sec period before the CS came on during every trial. The column labeled Freeze During CS tells how many units of freezing and other fear-related behaviors Sniffy produced during the CS. The column labeled Active During CS tells how many units of active, "non-frozen" behavior Sniffy produced during the CS. The column labeled Movement Ratio is the proportion of freeze units during

the CS: Movement Ratio = Freeze During CS / (Freeze During CS + Active During CS). Starting with Row 2, each row of data contains the information about a particular classical conditioning trial.

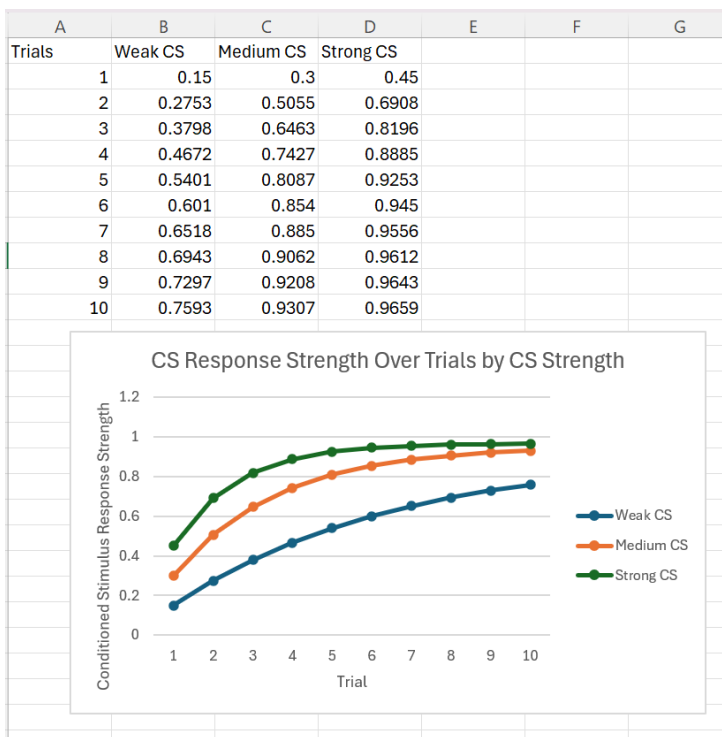
Analyzing the Data: Each of your three data export files contains information about a particular intensity of the tone CS. The data that we need to draw our graph showing the effect of CS intensity on CS response strength appears in the column labeled Tone in each of the three data export files. To draw a single graph in which we compare CS response strength as a function of trials for the three different CS intensities, we need to copy the tone data from the three data export files and paste those data into the spreadsheet file that we named CS Intensity. The columns on the CS Intensity spreadsheet are labeled Trials, Weak CS, Medium CS, and Strong CS. First, we will copy the data from the low-intensity tone data export file into the column called Weak CS on the spreadsheet. To do this:

1. **Go to the Low CS data export file.** Most spreadsheet programs have a pull-down menu called Windows. Your already open Low CS data file should be one of the menu choices under the Windows menu.
2. **Select the numbers in the column labeled Tone.** In most programs, you can select a group of numbers in a column by pointing the cursor at the top number, clicking your (left) mouse button and then keeping the button depressed while dragging down to the bottom number. When all the numbers have been selected, release the mouse button.
3. **Choose the Copy command,** which will probably be located under the Edit menu. Move back to the spreadsheet file into which you want to paste the numbers. To move back to the spreadsheet file, select it from the Windows menu.
4. **Select the cells for Trials 1 through 10 under the heading Weak CS.** To select these cells, click on the cell for Trial 1, drag down to the cell for Trial 10 while holding down the (left) mouse button, then release the mouse button. Select the Paste command from the Edit menu. The numbers that you copied from the Ex23-LoCS data export file should appear in the spreadsheet so that your spreadsheet now looks something like this:

Trials	Weak CS	Medium CS	Strong CS
1	0.15		
2	0.2752		
3	0.3798		
4	0.4671		
5	0.54		
6	0.6009		
7	0.6518		
8	0.6942		
9	0.7297		
10	0.7593		

5. **Repeat the same process for the medium and high intensity CS until all three columns are filled.** Now you are ready to draw your graph. From this point on, we can give only sketchy instructions because different spreadsheet programs, and even different versions of the same program, take somewhat different approaches to graph drawing. You will very likely have to read the manual and/or consult the online Help files for the program that you are using.

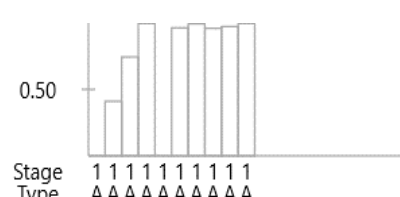
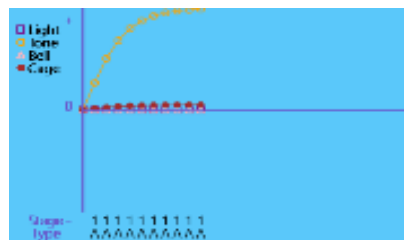
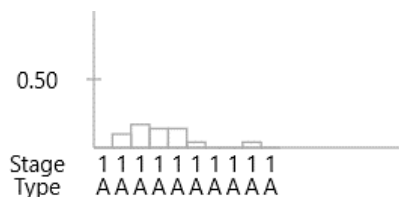
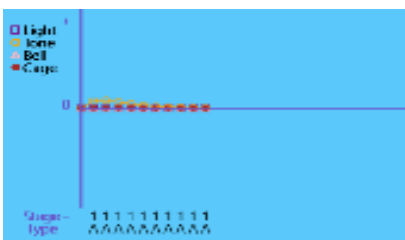
6. **The next step is to select the data that you want to include in the graph.** Some programs want you to select just the numeric data to be plotted (cells B2 through D11). Others want you also to select the column headers for the numeric data (cells B1 through D11), whereas still others want you to include the column labeled Trials (cells A1 through D11) as well.
7. **Once you have selected the right cells, you will choose a graph drawing command from one of the pull-down menus and then make a series of decisions about what kind of graph you want to draw.** These commands and choices vary considerably from program to program. In general, we can say that you want to draw a **line graph** and that you will want to create a **legend** to differentiate the different CS intensities. You will almost certainly need to consult the manual and/or the online Help to determine what to do. Eventually, your finished graph should resemble the following:



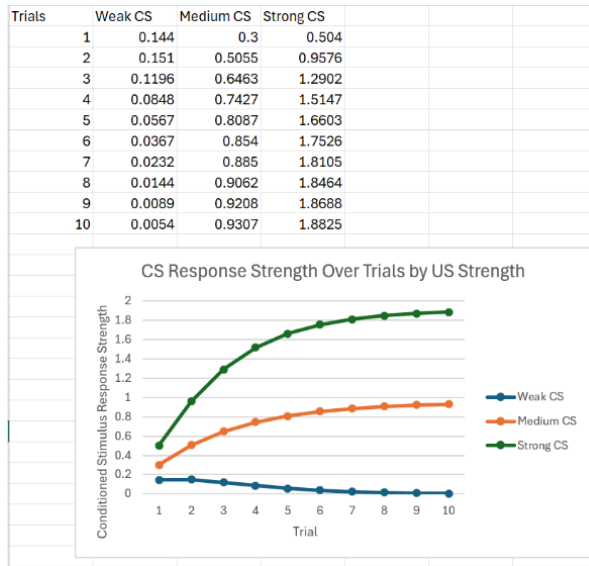
Running Exercise 5: Like the CS, the intensity of the US can also affect how quickly a rat can be classically conditioned. The stronger the US, the faster the rat learns and the greater the level of conditioning. It is also important to note that a low intensity US results in habituation of the UR while a high intensity US causes the UR to be sensitized. In the experiment, you will look at the differences between low, medium, and high intensity US.

1. **Open a new Sniffy file and immediately give it a new name. Save it in a Sniffy file folder on your computer (For example, ex4-lowtoneCS).**
2. **Under the experiment tab, click the ‘Design Classical Conditioning Experiment.’**
 - a. **Set the “Interval Between Trials” to 5 minutes.**
 - b. **Set the ‘Present Each Trial Type’ to 10 times.**
 - c. **In the ‘First Stimulus’ box select medium intensity tone.**
 - d. **In the “Second Stimulus’ box select either low or high intensity shock (you will do both).**
3. **Make sure the ‘Isolate’ button is clicked and the ‘Show After Experiment Completes’ is clicked as well, then push run and SAVE THE FILE.** Once the low intensity has been done, **repeat these steps to do the high intensity tone.** The medium intensity shock was done in exercise 1. **Next, you will need to make your own graph using excel.**

Example Graphs: (The left image is low intensity, and the right is high intensity).



4. Follow the same set of instructions listed in exercise 4 to make an excel graph of this data. The excel data and graph should look similar to this:



Task(s):

- **Submit** the document for Exercise 3 (*I will see Exercise 1 and 2 as well, since exercise 3 is based on having run Exercises 1 and 2*), exercises 4 and 5 and submit the Movement Ratio & CS Response Strength windows for Exercises 3, 4, & 5. **Make sure to include the line graphs that you made (as shown in the examples).**
- **Submit** your interpretation of your results.
- **View the video lecture** on the importance of classical conditioning.

Sniffy Projects 4-7 Instructions

In this assignment you will create a single document for all four projects you have completed with the Sniffy software. Please be sure to read all project instructions carefully and do not skip over sections, otherwise, your report may be incomplete.

Project 4 looks at latent inhibition, **Project 5** teaches about compound conditioning, **Project 6** assesses the sensory precondition and higher-order conditions, and **Project 7** explains excitatory and inhibitory conditioning.

- For Project 4 you will:
 - Submit the Movement Ratio and the CS Response Strength windows from exercises 7 and 8.
 - Interpret the Movement Ratio and CS Response Strength results

- Answer the following questions:
 - “How does pre-exposure to stimuli relate to learning in the real-world?”
 - “What, if any, differences occur with respect to pre-exposure to a CS or US?”
- For Project 5 you will:
 - Submit the Movement Ratio and the CS Response Strength windows from exercises 7 and 8.
 - Interpret the Movement Ratio and CS Response Strength results.
 - Your interpretation should be separate for each graph and very thorough if you want to receive full credit.
- For Project 6 you will:
 - Submit the Movement Ratio and CS Response Strength windows for Exercise 12 and 13.
 - You do not have to complete the unpaired control condition in Exercise 12.
 - Interpret the Movement Ratio and CS Response Strength windows.
 - Your interpretation should be separate for each graph and very thorough if you want to receive full credit.
- For Project 7 you will:
 - Submit the Movement Ratio and CS Response Strength windows for Exercise 10 and 11.
 - Interpret the Movement Ratio and CS Response Strength results.
 - Your interpretation should be separate for each graph and very thorough if you want to receive full credit.

Project 4: CS Pre-Exposure Effect (Exercise 20) and US Pre-Exposure Effect (Exercise 21)

Running Exercise 20: In this project, you will examine the effects of pre-exposure to the CS (*Exercise 20*) and pre-exposure to the US (*Exercise 21*). **Pre-exposure effect** is also known as **latent inhibition**. **Latent inhibition** is when a CS is given to an animal multiple times before that CS is paired with a US. Due to this prior exposure to the CS, the animal’s conditioning will take much longer. This can be explained in the chart below. (CS_{MT} stands for medium tone CS and US_M stands for medium US).

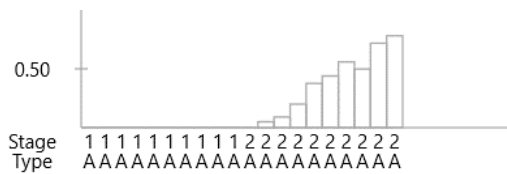
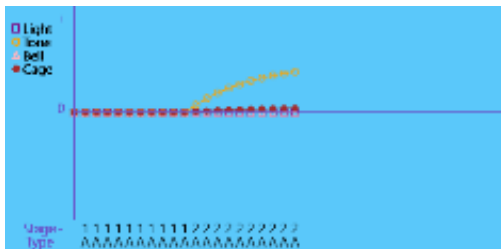
Condition	Stage 1	Stage 2	Stage 2 Expected Result
Experimental	10: CS _{MT} –None	10: CS _{MT} –US _M	Conditioning slower than usual
Control	Rest	10: CS _{MT} –US _M	Conditioning at normal rate

The chart shows that the experimental group received 10 trials of a medium intensity tone with no US while the control group received nothing. In the second stage, both groups received a medium intensity tone with a medium US. The experimental group learned slower than the control group due to the prior exposure. Follow the steps below to run your own experiment.

1. **Open a new Sniffy file and immediately give it a new name. Save it in a Sniffy file folder on your computer.**
2. **Under the experiment tab, click the ‘Design Classical Conditioning Experiment.’**
 - a. **Set the “Interval Between Trials’ to 5 minutes.**
 - b. **Set the ‘Present Each Trial Type’ to 10 times.**
 - c. **In the ‘First Stimulus’ box select medium intensity tone.**

- d. In the 'Second Stimulus' box select 'None.'
3. Before running this, click 'New Stage' and fill in the following:
 - a. Set the "Interval Between Trials' to 5 minutes.
 - b. Set the 'Present Each Trial Type' to 10 times.
 - c. In the 'First Stimulus' box select medium intensity tone.
 - d. In the "Second Stimulus' box select medium intensity shock.
 4. Make sure the 'Isolate' button is clicked and the 'Show After Experiment Completes' is clicked as well, then push run and SAVE THE FILE.

Example Graphs:

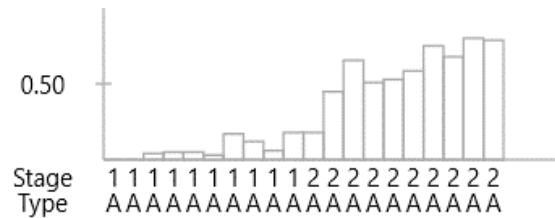
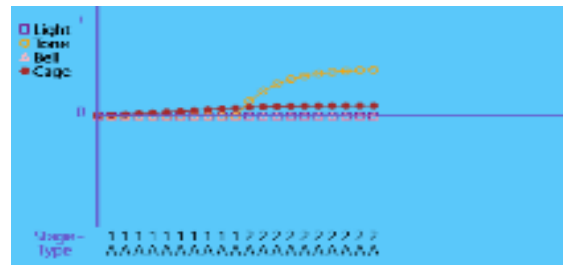
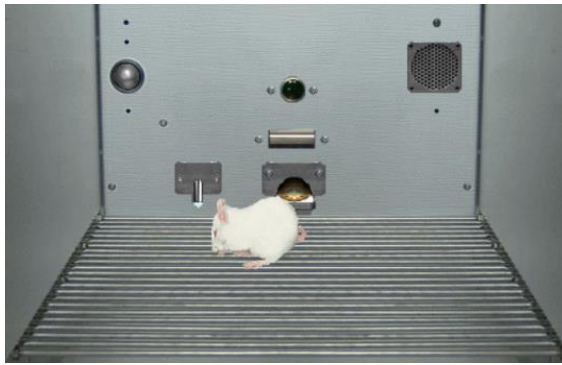


Running Exercise 21: This experiment dives into **US pre-exposure effects**; these are similar to CS pre-exposure effects but have US-only trials at the beginning instead of CS. The chart below illustrates that the experimental group received 10 trials of just a medium intensity US while the control received nothing. In the second stage, both groups received 10 trials of a medium intensity tone paired with a medium US. The experimental group will have less efficient conditioning than the control because they were exposed previously to the US. Follow the instructions below to complete your experiment.

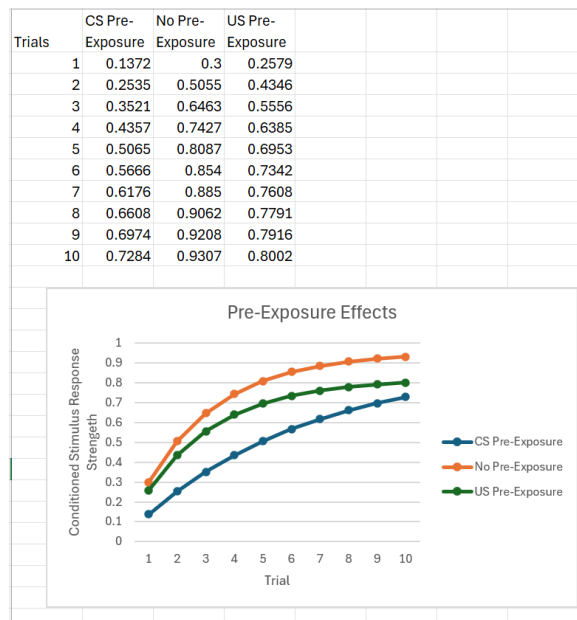
Condition	Stage 1	Stage 2	Stage 2 Expected Result
Experimental	10: US _M	10: CS _{MT} -US _M	Conditioning less efficient
Control	Rest	10: CS _{MT} -US _M	Conditioning normal

1. Open a new Sniffy file and immediately give it a new name. Save it in a Sniffy file folder on your computer.
2. Under the experiment tab, click the 'Design Classical Conditioning Experiment.'
 - a. Set the "Interval Between Trials" to 5 minutes.
 - b. Set the 'Present Each Trial Type' to 10 times.
 - c. In the 'First Stimulus' box make sure there is no CS selected.
 - d. In the 'Second Stimulus' box select medium intensity shock.
3. Before running this, click 'New Stage' and fill in the following:
 - a. Set the "Interval Between Trials" to 5 minutes.
 - b. Set the 'Present Each Trial Type' to 10 times.
 - c. In the 'First Stimulus' box select medium intensity tone.
 - d. In the "Second Stimulus" box select medium intensity shock.
4. Make sure the 'Isolate' button is clicked and the 'Show After Experiment Completes' is clicked as well, then push run and SAVE THE FILE.

Example Graphs:



5. **Compare the effects of US (exercise 21) and CS pre-exposure (exercise 20) using your own graph on excel.** Follow the same set of instructions **listed in Project 3, exercise 4** to make an excel graph of this data. This data will reflect the different pre-exposure effects on conditioning by using the data you just collected from Sniffy. The data from Project 3 Exercise 1 will serve as your no pre-exposure effects. The excel data and graph should look similar to this:



Task(s):

- **Submit** the Movement Ratio & CS Response Strength windows for Exercises 20 & 21.
- Submit the **graphs that you made** (as shown in the examples).

- Submit **your interpretation** of your results.
- **Answer the following questions and submit:**
- *“How does pre-exposure to stimuli relate to learning in the real-world?”*
- *“What, if any, differences occur with respect to pre-exposure to a CS or US?”*

Project 5: Blocking (Exercise 7) and Overshadowing (Exercise 8)

In these exercises, you will examine the effects of compound CSs and the relationship of their timing and presentation to the CR.

Running Exercise 7: This project moves into the realm of compound conditioning. **Compound conditioning** is when two or more CS’s are presented together. This model is more realistic, as many events use multiple senses at once, like sight and sound. Experiment seven looks at blocking, which is what happens when one of the components of the compounded CS has already been paired with the US enough times to cause a strong CR. The chart below illustrates how blocking occurs.

Condition	Stage 1	Stage 2	Stage 3 Expected Result
Blocking Experimental Condition	10: CS _{MT} –US _M	10: CS _{ML} &CS _{MT} –US _M	1: CS _{ML} –None CS _{ML} elicits little or no CR 1: CS _{MT} –None CS _{MT} elicits a strong CR
Compound Conditioning Control	Rest	10: CS _{ML} &CS _{MT} –US _M	1: CS _{ML} –None CS _{ML} elicits a moderate CR 1: CS _{MT} –None CS _{MT} elicits a moderate CR

In stage one, the blocking group receives 10 trials of medium intensity tone paired with a medium intensity US while the control group receives no trials. In stage two, both groups received 10 trials of medium intensity light and medium intensity tone paired with a medium intensity US. In the third stage, both groups received either the medium intensity light or the medium intensity tone with no paired US. In the blocking group, the light caused a minimal or no response while the tone, because it was already paired previously, elicited a strong CR. The control group showed a moderate CR to both light and tone because they had only ever been presented together.

Blocking Stage 1:

1. **Open the file from Experiment 3 exercise 1. Choose Design Classical Conditioning Experiment from the Experiment menu and verify that the following conditions were in effect for Stage 1:** The average Interval Between Trials was 5 minutes. The number below Present Each Trial Type was 10. The First Stimulus was the medium-intensity tone. The Second Stimulus was the medium-intensity shock US. Stage 1 is the only stage that has been run (i.e., the file that you are looking at is not the file in which Sniffy was subsequently extinguished in Stage 2). The file should have no stage except Stage 1. **If you have the right file, use the Save As command to save the file under an appropriate**

new name (e.g., Ex7-Blocking) in the Sniffy Files folder on your computer's hard drive and skip down to the next heading, "Blocking Stages 2 and 3." If you do not have your original conditioning file, go back to Exercise 1 and follow the instructions given there to create Stage 1 of the current experiment.

Blocking Stages 2 and 3:

- 2. Under the experiment tab, click the 'Design Classical Conditioning Experiment.' Then, click the 'New Stage' button to move onto the second stage.**
 - a. Set the "Interval Between Trials' to 5 minutes.**
 - b. Set the 'Present Each Trial Type' to 10 times.**
 - c. In the 'First Stimulus' box select BOTH medium intensity tone and medium intensity light.**
 - d. In the 'Second Stimulus' box select medium intensity shock.**
- 3. Before running this, click 'New Stage' and fill in the following (this will be labeled as Stage 3, Trial A):**
 - a. Set the "Interval Between Trials' to 5 minutes.**
 - b. Set the 'Present Each Trial Type' to 1 time.**
 - c. In the 'First Stimulus' box select medium intensity LIGHT only.**
 - d. In the "Second Stimulus' box select 'none.'**
- 4. Before running this, click the 'New Type' button under the 'Trial Types' section of the dialogue box. This should be labeled as Stage 3, Trial B.**
 - a. In the 'First Stimulus' box select medium intensity TONE only.**
 - b. In the "Second Stimulus' box select 'none.'**

Example Window:

does not increase very much. Stage 3 confirms what was seen in the CS Response Strength window in stage 2; the light does not elicit a fear response but the tone does.

Running Exercise 8: Overshadowing is a phenomenon that is looked at during a compound conditioning exercise where the components are of unequal salience. **Salience** means the stimulus' ability to attract attention. The chart below explains this phenomenon.

Condition	Stage 1	Stage 2	Stage 2 Expected Result
Overshadowing Experimental Treatment	10: CS _{HT} &CS _{LL} -US _M	1: CS _{ML} -None 1: CS _{MT} -None	CS _{ML} -Weak CR CS _{MT} -Strong CR
Compound Conditioning Control	10: CS _{MT} &CS _{ML} -US _M	1: CS _{MT} -None 1: CS _{ML} -None	CS _{MT} -Moderate CR CS _{ML} -Moderate CR

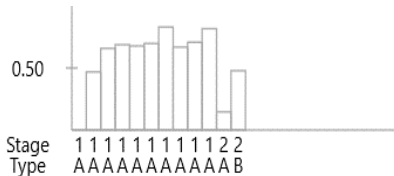
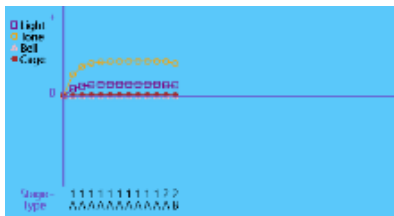
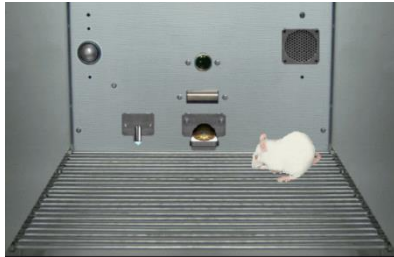
In stage 1, the experimental group receives a high intensity tone and a low intensity light with a medium US while the control group gets a medium intensity tone and medium intensity light with a medium intensity US. In stage 2, both groups get a medium intensity tone with no US and a medium intensity light with no US. The experimental group will have a weak CR to the light and a strong CR to the tone while the control group will have a moderate CR to both the light and the tone. This illustrates how differing saliences can affect the outcome of an experiment. Follow the instructions below.

1. **Open a new Sniffy file and immediately give it a new name.** Save it in a Sniffy file folder on your computer.
2. **Under the experiment tab, click the 'Design Classical Conditioning Experiment.'**
 - a. **Set the "Interval Between Trials" to 5 minutes.**
 - b. **Set the 'Present Each Trial Type' to 10 times.**
 - c. **In the 'First Stimulus' box select BOTH the low intensity light and the high intensity tone.**
 - d. **In the 'Second Stimulus' box select medium intensity shock.**
3. **Before running this, click 'New Stage' (Stage 2, Trial A) and fill in the following:**
 - a. **Set the "Interval Between Trials" to 5 minutes.**
 - b. **Set the 'Present Each Trial Type' to 1 time.**
 - c. **In the 'First Stimulus' box select medium intensity LIGHT only.**
 - d. **In the "Second Stimulus" box select 'None.'**
4. **Before running this, click the 'New Type' button under the 'Trial Types' section of the dialogue box. This should be labeled as Stage 3, Trial B.**

- a. In the 'First Stimulus' box select medium intensity TONE only.
- b. In the "Second Stimulus' box select 'none.'

5. Make sure the 'Isolate' button is clicked and the 'Show After Experiment Completes' is clicked as well, then push run and SAVE THE FILE.

Example Graphs:



Hint: When the salience of the compounded components differs, the CS Response strength will differ as well. Overshadowing has occurred because the high intensity CS developed a higher CS response strength than the low intensity CS.

Task(s):

- Submit the **Movement Ratio & CS Response Strength windows** for Exercises 7 & 8.
- Submit your **interpretation** of your results.

Project 6: Sensory Preconditioning (Exercise 12) and Higher-Order Conditioning (Exercise 13)

This project expands your understanding of the complex relationships that can occur as a result of classical conditioning. For example, the phenomena of sensory preconditioning and higher-order conditioning illustrate what happens when you pair two CSs. In Exercise 12, make certain that you also perform the unpaired control condition.

Terms: **First-order conditioning** is another name for conditioning where a CS is paired with a US to create a response. **Orienting response (OR)** is what occurs from a CS by itself and is hardly noticeable. **Sensory preconditioning** is when CS₁ and CS₂ are paired before a CR is conditioned to CS₂. **Higher-order conditioning** occurs when CS₁ and CS₂ are paired after a CR is conditioned to CS₂. CS₁ being the first CS and CS₂ being the second CS in a sequence. S-R association is the connection between the stimulus and the response.

Background: The main point of this unit is to demonstrate that learning can occur as a result of pairing two CS's. Earlier we talked about how a US can elicit an unconditioned response (UR); however, a CS also has a response- just not as noticeable. A CS initially causes an orienting response (OR) that is so unnoticeable that many scientists do not bother to measure it. If we pair a CS and a US, there will be an obvious response but if we pair two CS's how do we know if anything was learned? Sensory preconditioning and higher-order conditioning allow us to answer this question. The reason two CS's do not elicit an obvious response is because they both do not create a UR. The best way to get a CS to create an obvious response is to pair it with a US. It does not matter whether the CS's are paired before or after the pairing of CS₂ to the US. If paired before the US, it is called sensory preconditioning but if paired after the US then it is called higher-order conditioning. This is explained in the chart below.

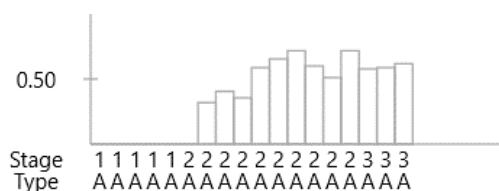
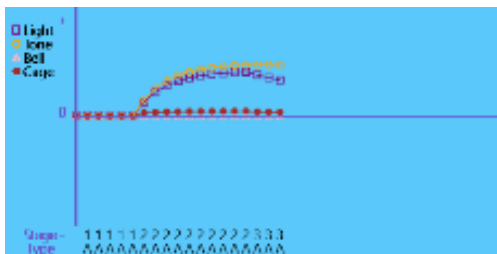
Condition	Stage 1	Stage 2	Stage 3	Stage 3 Expected Result
Experimental	5: CS _{ML} -CS _{MT}	10: CS _{MT} -US _M	3: CS _{ML} -None	CS _{ML} elicits a CR
Unpaired Control	5: CS _{ML} -None 5: CS _{MT} -None	10: CS _{MT} -US _M	3: CS _{ML} -None	CS _{ML} elicits No CR

In this unit we will be doing S-S associations, meaning that the first stimulus (CS) is paired with the second stimulus (CS). In previous units we have worked with S-R associations where the first stimulus (CS) is paired with a US. Follow the instructions below.

Running Exercise 12: This will demonstrate how to perform sensory preconditioning.

1. **Open a new Sniffy file and immediately give it a new name.** Save it in a Sniffy file folder on your computer.
2. **Under the experiment tab, click the 'Change Nature of Association.'**
 - a. **Select the S-S association for CS used as US and click OK.**
3. **Under the experiment tab, click the "Design a Classical Conditioning Experiment.'**
 - a. **Set the "Interval Between Trials' to 5 minutes.**
 - b. **Set the 'Present Each Trial Type' to 5 times.**
 - c. **In the 'First Stimulus' box select the medium intensity light.**
 - d. **In the 'Second Stimulus' box select medium intensity tone under CS used as US.**

- ### Example Graphs:



Running Exercise 13: This will demonstrate how to perform higher-order conditioning.

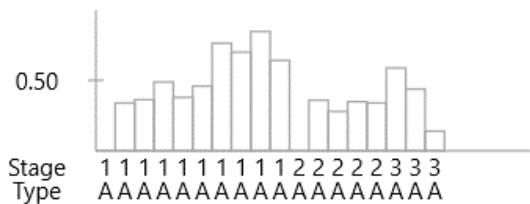
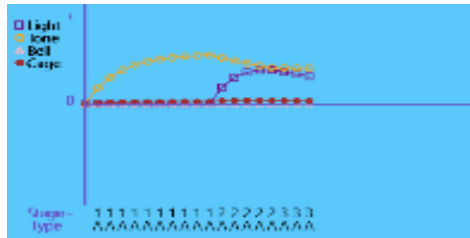
Stage 1:

1. **Open the file from Project 3 exercise 1. Choose Design Classical Conditioning Experiment from the Experiment menu and verify that the following conditions were in effect for Stage 1:** The average Interval Between Trials was 5 minutes. The number below Present Each Trial Type was 10. The First Stimulus was the medium-intensity tone. The Second Stimulus was the medium intensity shock US. Stage 1 is the only stage that has been run (i.e., the file that you are looking at is not the file in which Sniffy was subsequently extinguished in Stage 2). The file should have no stage except Stage 1. If you have the right file, use the Save As command to save the file under an appropriate new name (e.g., Ex13-higherorder) in the Sniffy Files folder on your computer's hard drive and skip down to the next heading, "Stages 2 and 3." If you do not have your original conditioning file, go back to Exercise 1 and follow the instructions given there to create Stage 1 of the current experiment.

Stage 2 and 3:

2. **Under the experiment tab, click the 'Design Classical Conditioning Experiment.' Then, click the 'New Stage' button to move onto the second stage.**
 - a. **Set the "Interval Between Trials' to 5 minutes.**
 - b. **Set the 'Present Each Trial Type' to 5 times.**
 - c. **In the 'First Stimulus' box select medium intensity light.**
 - d. **In the 'Second Stimulus' box select medium intensity tone as a CS used as US.**
3. **Before running this, click 'New Stage' and fill in the following (this will be labeled as Stage 3, Trial A):**
 - a. **Set the "Interval Between Trials' to 5 minutes.**
 - b. **Set the 'Present Each Trial Type' to 3 times.**
 - c. **In the 'First Stimulus' box select medium intensity LIGHT.**
 - d. **In the "Second Stimulus' box select 'none.'**
4. **Make sure the 'Isolate' button is clicked and the 'Show After Experiment Completes' is clicked as well, then push run and SAVE THE FILE.**

Example Graphs:



Hint: The movement ratio graph confirms what the chart above predicted was correct.

Task(s):

- **Submit the Movement Ratio & CS Response Strength windows for Exercises 12 & 13** (*you do not have to complete the unpaired control condition in Exercise 12*).
- Submit your **interpretation** of your results.

Project 7: Inhibitory Conditioning (Exercises 10 and 11)

Thus far, our classical conditioning has been excitatory only, which means that the CS predicts the occurrence of the US. However, equally important influences in learning are signals that indicate that something that might be expected will not occur. This is called inhibitory classical conditioning and it is the focus of these exercises.

Terms: Excitatory Conditioning is when the CS elicits a CR. **Inhibitory Conditioning** is when the CS “signals” that the predicted outcome of a response will not occur.

Background: Inhibitory CS do not elicit a CR when presented by themselves, after all if it predicts no reward will be given for a response. Inhibitory conditioning is thus often measured in conjunction with an excitatory CS. When this occurs, a well-trained inhibitory CS will slow/weaken the CR to the excitatory CS. By doing this you can slow the acquisition of a CR to an excitatory CS and reduce the CR to an already existing CS-CR pairing.

Running Exercise 10: To see if inhibitory conditioning is working you will need to run two different experimental conditions. The chart below visualized what you will be doing:

Condition	Stage 1	Stage 2	Stage 2 Expected Results
Experimental	10: CS _{ML} - US _M 10: CS _{MT} & CS _{ML} - None	10: CS _{ML} - US _M	Slower conditioning
Control	Rest	10: CS _{ML} - US _M	Faster conditioning

*In the above chart ML = Medium intensity Light; M = Medium intensity Shock; MT = Medium intensity Tone

To make your interpretation you will need to compare the graphs from the Experimental Condition and Control Condition. In stage 1 Trial Types A and B will be randomized. You should notice as the stage progresses the movement ratio will increase for Trial A but not Trial B. Trial Type B may either stage the same or decrease depending on the random order of your trials. In Stage 2 look at the CS Response Strength to determine if the inhibitory conditioning slowed the acquisition of a fear response to the shock compared to the control condition.

Experimental Condition:

1. **Open a new Sniffy file and immediately give it a new name.** Save it in a Sniffy file folder on your computer.
2. **Under the experiment tab, click the ‘Design Classical Conditioning Experiment.’**
 - a. **Verify you are editing Stage 1, Trial Type A**
 - i. **Set ‘Interval Between Trials’ to 5 minutes**
 - ii. **Set ‘Present Each Trial Type’ to 10**
 - iii. **Set ‘First Stimulus’ to medium-intensity light**
 - iv. **Set ‘Second Stimulus’ to medium-intensity shock US**
 - b. **Click New Type**
 - c. **Verify you are now editing Stage 1, Trial Type B**
 - i. **Set ‘First Stimulus’ to *both* medium-intensity light and medium-intensity tone**
 - ii. **Set ‘Second Stimulus’ to none**
 - d. **Click New Stage to move to Stage 2**

e. Verify you are editing Stage 2, Trial Type A

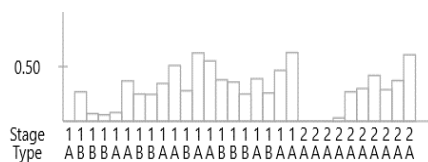
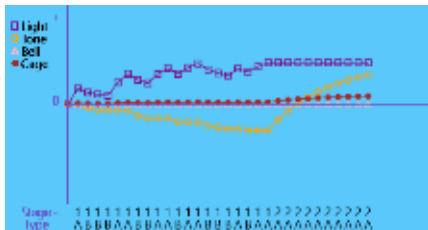
- i. Set 'Interval Between Trials' to 5 minutes
- ii. Set 'Present Each Trial Type' to 10
- iii. Set 'First Stimulus' to medium-intensity tone
- iv. Set 'Second Stimulus' to medium-intensity shock US

3. After checking and verifying all settings click Run and then SAVE the file.

Control Condition:

- a. The control condition should be the exact same as exercise 1. You should use that exercise to compare as your control.

Example Graphs:



Running Exercise 11: In this experiment we will demonstrate how an inhibitory CS elicits weaker CR to a different excitatory CS. The chart below will visualize the experimental design:

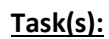
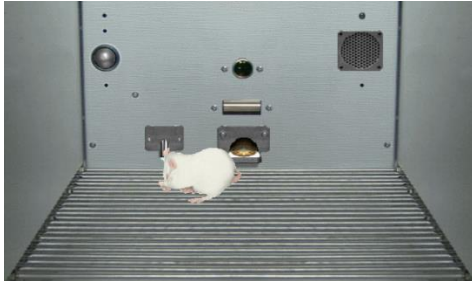
Stage 1	Stage 2	Stage 2 Expected Result
10: CS _{ML} - US _M	1: CS _B - None	CS _B elicits strong CER
10: CS _{MT} & CS _{ML} - None	1: CS _{MT} & CS _B - None	CS _{MT} & CS _B elicits weak CER
10: CS _B - US _M		

In the above chart ML = Medium intensity Light; M = Medium intensity Shock; MT = Medium intensity Tone; CS_B = Bell CS; CER = Conditioned Excitatory Response

For this exercise the first 2 Trials for Stage 1 will be the same as what we ran for Exercise 10. The third Trial will train a CR to the bell CS. In Stage 2 we present the bell alone to make sure it elicits a CR in Trial 1. Then in Trial 2 we are looking to see if the medium-intensity tone will suppress the CR of the bell. You will check the Movement Ratio and CS Response Strength windows to make these determinations.

1. **Open a new Sniffy file and immediately give it a new name.** Save it in a Sniffy file folder on your computer.
2. **Under the experiment tab, click the 'Design Classical Conditioning Experiment.'**
 - a. **Verify you are editing Stage 1, Trial Type A**
 - i. **Set 'Average Interval Between Trials' to 5 minutes**
 - ii. **Set 'Present Each Trial Type' to 10**
 - iii. **Set 'First Stimulus' to medium-intensity light**
 - iv. **Set 'Second Stimulus' to medium-intensity shock US**
 - b. **Click New Type**
 - c. **Verify you are now on Stage 1 Trial, Type B**
 - i. **Set 'First Stimulus' to medium-intensity light and medium-intensity tone**
 - ii. **Set 'Second Stimulus' to none**
 - d. **Click New Type**
 - e. **Verify you are now editing Stage 1, Trial Type C**
 - i. **Set 'First Stimulus' to bell**
 - ii. **Set 'Second Stimulus' to medium-intensity shock US**
 - f. **Click New Stage to move to Stage 2**
 - g. **Verify you are now editing Stage 2, Trial Type A**
 - i. **Set 'Average Interval Between Trials' to 5 minutes**
 - ii. **Set 'Present Each Trial Type' to 1**
 - iii. **Set 'First Stimulus' to bell**
 - iv. **Set 'Second Stimulus' to none**

- ### Example Graphs:



- 43 | Page

Phase II: Operant Conditioning with a Live Animal

Lab Facility: For phase II, the operant conditioning chambers and your rat are located in the Baylor Sciences Building (BSB), Room A323.

- The rats are kept in a special room inside Room A323, which is called a colony room.
- Each rat is individually housed in a plastic cage.
- Your rat requires water every day and is fed a laboratory "chow" – a pelletized, high-protein food specifically designed to provide balanced nutrition.
- Bedding material covers the bottom of their cages to collect your rat's wastes. This bedding is changed on a regular basis by a full-time animal care technician.
- The cages in which the animals live are also cleaned regularly, and these sanitation practices ensure that the colony is a healthy place for your rat.
- As is true of most life forms, rats prefer alternating cycles of day (*lights on*) and night (*lights off*). Therefore, most colonies (*including ours*) are kept on a 12-hour on, 12-hour off cycle that matches the outside world.

NOTE: Baylor University extends to students the privilege of working with live animals. All rats in this lab belong to Baylor University. Rats or other animals are not allowed to be introduced into or taken from this facility, and Baylor may take disciplinary action against individuals who do not comply with laboratory rules.

General Laboratory Procedures and Policies: You are about to enter a new realm. The sub-culture of laboratories is more easily learned if you know some rules, conventions, and procedures before entering the laboratory. Some of these rules have an empirical base – that is, we use procedures that have been developed by hundreds of laboratories over many decades and which comply with federal regulations. For a variety of reasons, rats and mice are often the experimental animals of choice, and most lab rats are an albino strain of native rat called *rattus norvegicus* (*Norway rat*). In this lab, when you start working with a live animal, you will be working with a male Sprague-Dawley rat. To help you overcome your hesitancy in handling them, and, more importantly, to overcome the rat's fear of you, we have scheduled a "getting acquainted" session before formal experimentation begins (*see calendar*).

Lab Environment:

In our lab, the work environment is "open" to allow you to observe your rat's every movement. Unfortunately, this also allows your rat to see and hear you and the other experimenters in the room. For this reason, you must refrain from all unnecessary talking, you must keep your voice down, and you must not make any other unnecessary noise.

You must wear a lab coat at all times in this lab. There are no exceptions to this policy. It may seem like a waste of time to put on a lab coat each time you enter the lab, but it is not. Lab coats not only keep your clothes and the lab environment clean, they also help mask any cologne or perfume you are wearing, and help prevent possible injuries to the rats because of your rings. Extra lab coats are stored in the colony room, but you may also purchase one of your own if you want. We also have respiratory masks for you to wear, which might help prevent any allergic responses that you either already have or could eventually develop to these animals. Smoking, eating, and drinking are prohibited in the lab.

In the past, students have recognized that their rat does well when they – the student – sits still. Many students have stated *"Every time I pick up my pen to record data, my rat stops what it is doing and looks at*

me". This is a *startle reflex*, and for this reason, I suggest that you sit off to the side of the Skinner box when your rat is in it and hold your pen in your hand. If you sit still, your rat is more likely to cooperate with the experiment by acting naturally.

It is imperative that you use the same Skinner box every day! This is necessary because the number of the box identifies your data in the computer's output for every lab. You and your partner must also use the same operant conditioning chamber.

As previously stated, you are not allowed to do any of the operant conditioning experiments until you have carefully read each lab's experiment and you have taken a written quiz on the material for that particular lab session. Therefore, the quiz will be administered at the beginning of each lab session, before you obtain your rat and before the experiment begins.

Other Important Issues: The following are more important issues and apply to every lab session:

- Always bring your lab manual to every lab session, as you must record data into it and it will be periodically graded by the lab session instructor.
- This is a scientific laboratory, so you must:
 - Wear a lab coat and gloves.
 - Keep quiet!!!
 - Be observant, methodical, precise, and consistent!!!
- Quizzes are administered at the beginning of each operant conditioning lab session.
- See the calendar to identify the specific experiment that each lab covers.
- On the lab quizzes, don't over-think answers – they're all straight from this manual and from your course and lab textbooks.
- Turn the water bottles around when opening your rat's cage.
- For all operant conditioning lab sessions: Wipe down the Skinner box with the cleanser provided and clean up around your area before leaving the lab.
- For all operant conditioning experiments, you must wash your hands before leaving the lab.

Entering the lab and getting your rat:

- First, put on your lab coat and gloves.
- The rat's startle reflex typically elicits a freezing response to novel noises and moving figures, so you must always enter the colony room quietly.
- Approach the rack containing the cages, turn the water bottle around, and open your rat's cage carefully and quietly.
- Place the cage on the counter in the colony room.
- Remove your rat and weigh it. Immediately record its weight in both the colony logbook and in your lab manual (*additional information on record-keeping appears below*).

Handling your Rat:

- Reach into the rat's cage with your hand and pick up the rat as you would a cat. Attempt to wrap its whole body with your hand – never grab its leg, tail, or head.
- Your rat will be calm if you hold it gently but firmly in your hands and against your warm lab coat.
- Let your rat get comfortable in your hands. Rats are olfactory animals and they will learn the scent of their handlers very quickly. By simple association, if you do not treat your rat well, it will learn that your

scent is an unpleasant one and it will use its only defense against you – biting. Though it generally does not hurt when the rat bites you, it is much more pleasant to experiment with a rat that likes you. Treat your rat as you would a pet. It will not bite if you don't give it a reason to.

- Sweet-talk your rat, but keep it off your shoulders, away from your face (*especially those biting rats*), and away from the other exposed areas of your body.
- It is important that you and your lab partner both visit and handle your rat several times before you begin experimentation. The point of this important time period is for you to habituate to handling your rat and for your rat to habituate to being handled.
- Always wash your hands after handling your rat.

Colony logbook, weighing and watering your rat, access to the data and other record-keeping:

- A logbook is kept in the colony room on the counter by the sink. This logbook helps the director of the colony keep track of your compliance with the American Standards for the Care of Laboratory Animals, so it includes the names of students who work with these animals, the rats' names and numbers, the times and dates you work with your rat, and some basic information about the life that you are being entrusted with.
- The colony logbook is also used to record your attendance. If you attend the lab but forget to fill out the logbook, you will be counted as absent. There are no exceptions to this policy.
- It is important to sign the log book every day because the Baylor Animal Care and Use Committee checks in often to be certain that these animals are being properly cared for. If you fail to record your animal's weight and water consumption, the committee will decide that these animals are not being properly cared for and the lab will be shut down. If this occurs, it will SIGNIFICANTLY affect your grade for this lab.
- Weigh your rat daily and record this weight in both the colony logbook and in your lab manual.
- When you begin Experiment 1, you will weigh your animal and record its weight in the "Initial Weight" space on the inside front cover of your lab manual. It is very important to calculate 80% of the rat's initial weight and to note this value in the "80% Weight" space at the bottom of this same page. You will also record this information in the colony logbook and on the cage information card. You will also weigh your rat at the beginning of each subsequent lab session. Remember the "80% Weight" value every time you weigh your rat, and if your rat's weight drops below 80% of their initial weight, you must notify your lab instructor.
- At the end of every lab session, you must water your rat for exactly 10 minutes and record how much they drink in both the colony logbook and also in your lab manual.
- *Use the stopwatch or your wristwatch to make certain that your rat drinks for exactly 10 minutes. In science, methodology and strict attention to detail are very important.*
- For all operant conditioning experiments, your lab session instructor will upload the data from the computer onto Canvas as you will need this data to complete your assignments.
- In addition, you should also record your observations regarding how your animal is responding to your handling. Is your rat becoming more calm and gentle? Are you becoming more calm and gentle?

Your Responsibilities:

Working with a live rat is a privilege that you have been granted in this lab, and you will lose this privilege if you jeopardize the well-being of your lab animal, which includes being held responsible and accountable for feeding and watering your rat. There are no excuses for a person who abuses any living creature entrusted to your care. If your animal becomes ill, dies of neglect, or is in any way abused while in your care, you can rest assured that both you and your lab partner will fail this laboratory. Additionally, if you see an animal that looks like it is in poor condition, you must notify your laboratory instructor and your professor.

Daily Lab Activities: During the operant conditioning phase, this is generally how the labs will go:

- Wash your hands.
- Don your lab coat and gloves. Put on your mask if you choose to wear one.
- Take the quiz, if applicable.
- Get your rat from the colony room.
- Weigh your rat.
- Sign the logbook and record your rat's weight.
- Also record your rat's weight in your lab manual.
- Return to the operant conditioning chambers and put your rat into your chamber **(CAREFULLY! Do not close the door on the rat's tail).**
- The Lab TA will then begin to run the experiment.
- While the experiment is running, carefully observe your rat's behavior.
- After the experiment is over, remove your rat from the chamber, put it back into its home cage, and then water your rat for 10 minutes exactly.
- While your rat is drinking, thoroughly clean your chamber by removing the lower tray and washing it, and then wipe down the bars and the glass of the chamber with the EtOH-H₂O solution provided.
- Record the amount of water that your rat consumed in both the colony logbook and in your lab manual.
- Clean the Skinner box and the area around your work station.
- Wash your hands.
- If applicable, download the data for you and your partner from Canvas.
- Do the lab assignment and submit.

A Few Frequently Asked Questions

Question: Do these animals experience any pain or discomfort during these experiments?

Answer: We do not allow them to experience pain (*this is why we use positive reinforcement only in this course*). Additionally, we have taken special care to minimize any discomfort they might experience (*this is why you must attend and alert your lab session instructor about any absence*). Pain and discomfort to these animals should always be minimized.

Question: Isn't it cruel to make an animal live in a cage?

Answer: No. Baylor University houses and maintains experimental animals according to guidelines established by input from many researchers and the Federal government. Reasonable people may disagree with these guidelines. Again, we believe that experimentation is a worthwhile endeavor, and that rats housed in clean cages, with unlimited food, and sufficient access to water, in a temperature-controlled room, are being treated reasonably well (*or at least better than Mother Nature would treat them*).

Question: Does a veterinarian look after the rats?

Answer: Yes. Baylor University has a contract with a local veterinarian who sits on a committee that oversees the laboratory.

Question: What happens to these rats when the lab is finished?

Answer: These rats belong to Baylor University and they may be used by other Baylor researchers for other research or teaching purposes or breeding. However, if the rats are not needed for other Baylor identified purposes at the end of the course, the following order of disposition will be used:

1. Adoption as pets per Institutional and IACUC policy (see protocol in Appendix 4.0 on page 78 of this manual)
2. Other uses identified by Baylor and acceptable under local, state, and federal regulations
3. Humane euthanasia per the AVMA Guidelines of the Euthanasia of Animals and as approved by the IACUC.

Question: Can my partner and I submit identical assignments for the operant conditioning phase?

Answer: NO. All of these assignments are to be done INDIVIDUALLY even though you and your partner will be using the same data. You will not receive any credit for a copied assignment.

Experiment 1: Getting to Know Your Rat (*Handling*)

From this point on in this course, you will work with a live animal in an operant conditioning chamber (*i.e.*, a “Skinner box”). The sole purpose of this first lab session in this new environment is simply to acquaint you with your rat and to better familiarize you with the lab’s protocols during this phase of the course.

Important Reminder: The back cover of this manual is used as a running log of your rat’s weight and water consumed. Therefore, it is imperative that you record this information in this lab manual and in the colony logbook every day because you must record your rat’s weight and water consumption.

Skinner's research strategy focused on the effects of rewards and punishments on on-going behavior. A Skinner box environment allows you to control many visual, auditory, tactile, reinforcement, and punishment cues (*we will only use visual, auditory, tactile, and positive reinforcement in this course*). These stimuli eventually become the context against which the presence of the lever, the animal's responses to it, and the introduction of the reinforcer, stand out.

The Skinner box environment: The Skinner box in which you will be placing your rat is a small chamber consisting of four walls, a ceiling, and a grid floor. Out of one wall protrudes a lever, which is properly called a **manipulandum**. Eventually, your rat will be shaped to press this lever to receive positive reinforcement (*water*). That is, in many of the following experiments, when the rat depresses this lever, the computer is programmed to deliver water reinforcement. The presence of the lever and the delivery of reinforcers are the primary *independent variables*, and the Skinner box environment allows you to measure the following *dependent variables*:

- Rate of lever pressing
- Pattern of behavior under different conditions of reinforcement
- Drinking behavior

The observant reader will notice that this selection of dependent variables does not allow us to estimate the rat's intelligence or intelligence quotient (IQ). What do we risk in assuming that a rat has an IQ?

Your rat brings to the Skinner box innate patterns of foraging. That is, since it is quite thirsty, it is motivated to explore the environment. However, to say that it is "searching for water" as it pokes its nose into every corner would be anthropomorphic and probably violates Morgan's canon. However, your rat will thoroughly explore this novel environment (*or at least more so than would a well-fed, well-watered rat*). Therefore, you should not be surprised if your rat initially explores the operant conditioning chamber. However, as you will see, this exploratory behavior will diminish as your rat is “conditioned” by these experiments.

Your rat's life history: Your rat has been in a restricted environment for its entire life. It was born in a plastic breeding cage where it lived for 3-4 weeks with its mother (*a female rat with pups is called a dam*) and with its mixed-sex littermates. It was then removed to a "home cage" where it lived with a same-sex littermate until it graduated to the single isolation cage that it now occupies. Food and water have always been delivered to its home cage. Therefore, your rat has had limited experiences, which helps to make a more optimum learning situation.

Inspiring Your Rat to Work for You

Since you're now working with a live animal in an operant conditioning chamber (*i.e.*, a "Skinner box"), you will eventually want to be able to put your rat into the Skinner box and see a lever-pressing response in that environment. But before we can do that, we must restrict the rat's water intake so that its foraging behavior is elicited and we must also conduct magazine training. However, before we can even do that, we must get your rat used to you and to being handled, and we must get you familiar with the lab environment and its protocols. These are the goals for this first operant conditioning lab session.

Initially, your rat has about as much intrinsic interest in lever pressing as you do in reading the "*suggestions for further reading*" that follow a chapter that you know you won't be tested on. Remember that your rat is already living in solitary confinement and has a brain about the size of the tip of your little finger. Therefore, providing some motivation (*e.g.*, *water deprivation*) for the rat to work is elementary and mandatory. Conceptually, the contractual arrangement you will eventually reach with your rat (*at least from the rat's point of view*) looks something like this:

(If you) lever-press --> (you get to) drink --> (and, therefore,
you may continue to) live (in solitary confinement).

(If you) don't lever-press --> (you) don't drink --> (and you) die.¹

As you can see, you've got your rat by its whiskers. Controlling the rat's access to water does several things: (1) because rats do not eat nearly as much of their high protein lab chow when they are water deprived, they are hungry as well as thirsty; (2) hungry and thirsty rats will move around in a novel environment (*like the Skinner box*) foraging for food and water, and their discovery of water in the Skinner box will motivate them to make responses that you can shape.

How much water deprivation is necessary for optimum learning to occur? Skinner and other behavioral researchers suggested that laboratory animals should be food/water deprived to no more than 80% of their normal body weight to ensure a high level of motivation and optimum performance. Therefore, you must record your animal's weight daily and be aware of any weight changes that drop below 80% of its initial value (*see "Important Note" below*). **If this occurs, you MUST notify your lab instructor.**

Rats do very well, health-wise and performance-wise, if they are allowed 10 minutes access to water in their home cage at the end of each lab session. Use the stopwatch to precisely time your animal's drinking and to time their responses in the Skinner box.

Rats become quite proficient at licking the tube of their water bottle under these deprivation conditions, and will usually consume approximately 10 mL of water during a 10 minute period. During the course of this lab, it is not unusual for your rat to gain weight given this amount of water and the extra water they will get on the weekends (*and if they are ordered as juveniles, they are still growing as well*). Likewise, don't be surprised if they drop 10 or so grams in body weight during their first few days of water deprivation. Also, do not become alarmed if your rat does not drink much during their first 10 minute period (*i.e.*, *this lab session*). Your rat will be more water deprived, and thusly more motivated during the next lab session

¹ A little bit of humor here, folks. We are not going to abuse your animal.

(i.e., your partner's orientation lab session). However, as previously stated, if your rat's weight drops below 80% of its initial value, you must notify your lab instructor.

IMPORTANT NOTE! On this first day, you will weigh your rat. This is your rat's initial weight PRIOR to water deprivation. Use this number to determine how much weight your rat can lose before you should become alarmed. Record this weight on the inside front cover of this lab manual, colony logbook, and on the information card on your rat's cage. Remember this value every time you record your rat's weight, and if it drops below this value you must immediately notify your lab instructor.

Analyzing the relationship between you and your rat (*the rat's perspective*): Your success in training a rat to lever press depends upon a number of factors other than "thirst motivation." Recall that behaviorists do not like terms – psychological constructs – that refer to a rat's – or even to a human's – unobservable mental activities. (*Indeed, thirst is such a term. Hence, the operational definition of thirst is specified as "hours of water deprivation"*). To say then that rats develop expectancies about their environment violates Morgan's Canon (*look it up if you don't know what it means*). Rats are active because they are foraging for water to counteract the many hours of water deprivation. We should not say that water-deprived rats are "active because they are thirsty" because we cannot know exactly what is going on in a rat's mind. Therefore, from a behavioral science perspective, a rat's "expectancies" are better understood and described as conditioned responses. Nonetheless, rats do develop expectancies about their environment, which now includes you, the lab environment, and these training sessions. Therefore, you must develop consistent and predictable laboratory routines.

The most important routine is to train your rat at the same time each day, at least four days each week.

Given your busy schedule, you may wonder why it is necessary to work with your rat at the same time each day. In the first place, rats have circadian rhythms – that is, they make physiological adjustments to the light/dark cycles of the laboratory. Because rats are nocturnal animals, working with them during the day finds them at the least active part of their rest/activity cycle. Hunger and thirst will increase their activity levels above what they would be if they were granted *ad libitum* (*free access*) access to food and water.

By training your rat at the same time each day, it will develop expectancies of when it receives water and food. It will arouse itself from its low point in the activity cycle and become more active just prior to your arrival at the lab. From your perspective, your rat will be ready to work for you in anticipation of drinking and eating. (*Note that the "time of day" then becomes a conditioned stimulus for the rat's conditioned responses*).

Maintaining constant thirst levels: Another reason (*not unrelated to the first*) to train your rat at the same time each day is that you can also be assured that the animal is approximately 23 hours water deprived. That is, the animal's performance will be more predictable and consistent when it is about 23 hours deprived than if it was 12 hours or 30 hours "thirsty."

The motivation level (*thirst level*) will be close to maximal when the rat is 23 hours water deprived.

You are now becoming part of your rat's routine: Dogs and cats love the people who feed them. The presence of the "feeder" is a secondary reinforcer that has discriminative stimulus properties (*i.e., the person who does the feeding becomes a "S+" to the dog or cat*).

Rats may or may not develop similar expectations regarding the person who feeds them. They probably cannot visually recognize you as your cat does, but they certainly do recognize your smell when you handle them and they assuredly do develop expectations. Your presence signals an upcoming sequence of handling, weighing, being put into the Skinner box, and drinking, respectively. Make sure you do not miss out on this integral “relationship building” part of the lab. Handle your rat for a few minutes every day so they get used to you and you get used to them.

Why your rat prefers the same daily routine: Rats prefer behavioral homeostasis in much the same way that their physiology demands regularity of function around a reference level (*e.g., body temperature is regulated around a reference level by homeostatic mechanisms*). In similar fashion, rats develop expectancies about how they will be handled by humans.

Rat expectancies likely do not match human expectancies in every respect. For example, humans may not be the only sensation seeking animals, but to my knowledge, there are no observations suggesting that rats appreciate being flown through the air, that they prefer music to quiet, that they would rather be squeezed tightly rather than being held gently and firmly, or that they enjoy being dropped on the floor. As the rat's trainer, you become an expected part of the rat's daily routine, and deviations from that routine – such as being dropped on the floor or being exposed to excessive noise – will show in the rat's lever-pressing performance.

For all of these reasons, from the animal's perspective, one trainer is better than two. So, when you share training responsibilities with a lab partner, the two of you introduce more behavioral variability and therefore have less behavioral control than if only one of you worked with your rat. Therefore, the more closely that your handling of the rat matches that of your lab partner, the more behavioral control you will eventually have over your rat.

If possible then, it is best if you and your partner go together to this lab session in order to ensure that you are using similar training techniques. Later, if problems occur, having observed each other's handling techniques and training methods, might give you some insight into why you and your lab partner are getting different training results.

Communicating closely with your lab partner is critical. If you and your partner cannot get in touch after each lab session to discuss your rat's progress, leave a short note on the white board in the lab for your lab partner explaining the day's reinforcement schedule, response frequency, behavior, and any confounding variables that might have occurred. It is very important to be in communication with your partner concerning your rat's training progress.

Why your rat prefers to be trained for four days each week: Actually, it doesn't. It would much rather be eating, sleeping, or breeding. But if you run your animal four days each week at the same time each day, the rat's expectancy/physiology will generate better performance than if they were run *periodically*. We know this empirically, not intuitively. You may also be interested to know that if their research involves working with live animals, graduate students in behavioral science must often work with their animals every day of the week.

Lab Tasks:

- Handle your rat (*see the introduction section of this lab manual for more information*).
- Weigh your rat, and record your rat's weight in the colony logbook, in your lab manual, and on the cage information card.
- At the end of this lab session, give your rat access to water for exactly 10 minutes. Record the amount of water that your rat drank in your lab manual and in the colony logbook.
- Cage Information Card: Fill out the information card on the front of the cage with your name, your partner's name the name of the rat, the rat's initial weight, and the 80% value of that initial weight

Additional Notes:

- Take note of which Skinner box you used and let your partner know so they can use the same one.
- Leave messages for your partner on the white board in the lab, if necessary.

Experiment 2: Magazine Training

Goal of Operant Conditioning: Pavlov studied associative processes in dogs by measuring the conditioned reflex of salivation. Skinner showed us how rats could be trained to lever press and how this operant response changed due to the application of reinforcers and punishers (*Skinner, 1953; 1959*).

As you may have already observed, your rat's exploratory behavior appears to be both voluntary and purposive – that is, it seems to know what it is doing. By contrast, salivation in response to food is reflexive. For this reason, Skinner argued that operant behavior is emitted rather than elicited. Indeed, Skinner said that an emitted behavior is any instrumental response that operates on the environment. Further, every emitted behavior has a baseline or operant level of occurrence. For example, upon first entering the box, the rat's likelihood of sniffing the floor and rearing to "inspect" the walls is high, while depressing the lever has a low operant level. Therefore, the rat's lever pressing behavior must be molded through a process called shaping. However, before you can begin shaping your rat, you must first accomplish magazine training.

Magazine Training: In this experiment, you will initiate the first phase of this magazine training (*magazine is a military term for where provisions are kept*). Magazine training introduces the rat to the location of and to the sounds that are associated with positive reinforcement – in our case, water. That is, in this session, we are trying to get your now water-deprived rat to focus its attention to the end of the operant conditioning chamber where water is being delivered every 3 seconds. Since some specific lights and sounds accompany these deliveries of water reinforcement, your rat will start associating these cues with the water. Therefore, the whole purpose of this session is to simply get your now water-deprived rat to attend to the area of the operant conditioning chamber where water is being presented.

Magazine training is therefore accomplished by Pavlovian conditioning. The rat will associate the sound (CS_1) of the dispenser mechanism, the sight (*or position of the lever*, CS_2) and the feel (CS_3) of the dispenser with the water (US).

After drinking, your rat will likely groom for a while and perhaps explore the Skinner box. It may also be consolidating (*a nice cognitive term*) about what it has learned at this point:

- Where the water is located (*right next to the lever*).
- The sights and sounds of the electrically-activated water dispenser and the feel of the water licked from the magazine (*all CSs*).
- With continuing trials, CS_2 and CS_3 become secondary reinforcers via higher-order conditioning.

When magazine training has been accomplished, you will be ready to begin shaping your animal's behavior in your next session. Typically, magazine training can be accomplished within a single session, but just to make sure, we have scheduled two sessions (*one for each partner*).

The point of magazine training is to arrange for your rat to have a positive history of receiving water in a particular place in the Skinner box, and hopefully, for it to associate the sound of the water dispenser with the delivery of water.

Experiment 3: Shaping

Magazine training served the dual purpose of informing the rat where the water is and also reinforcing the rat for being in that particular area within the Skinner box. However, even after magazine training has been accomplished, the rat still doesn't know that its task is to respond on the lever. In this experiment, you will begin to shape the rat to accomplish this target behavior (*i.e., lever pressing*).

The importance of this experiment to the success of the other experiments, and the importance of your role in this shaping process, cannot be overstated. To properly shape your rat, you must be a good scientist. But what does that mean, precisely? It means that you must be quiet and very observant; you must be consistent and patient; you must both deliver reinforcement and withhold it when appropriate; and you must be very purposive and systematic (*read the pages on shaping by successive approximations in your course textbook or in some other source*).

Shaping is accomplished by operant conditioning. The rat is conditioned to engage in a behavior (*lever pressing*) that is novel to their repertoire of genetically influenced behaviors. Note the difference between this process and magazine training.

Before you begin to shape the rat in the laboratory, practice shaping a rat's behavior using the Sniffy program.

Start Shaping: Place your rat in the box and observe where it goes. Allow the rat to discover the water. Activate the dispenser by using the shaping switch a few times while your rat is drinking. If the rat doesn't immediately go to the water magazine, allow it to roam without reinforcement. The first stage of shaping is simply to reinforce the rat for being in the area of the box where the lever is located. Should the rat move toward the lever from another part of the cage, activate the water dispenser.

The sound of the water dispenser is a secondary reinforcer, which was established during magazine training. Arriving and drinking near the lever is the behavior you are reinforcing with both the secondary reinforcer (*sound*) and primary reinforcer (*water*). Because the boxes are programmed for continuous reinforcement (*CRF*) in this experiment, if the rat depresses the lever – even accidentally – it will receive reinforcement.

Can We Please Get Some Cooperation Here?: If your rat camps out in the back part of the box, it may be necessary to reinforce the rat for merely turning its head in the direction of the lever or for making any movement toward the lever. That is, for some rats, their orientation behavior must be reinforced by delivery of positive reinforcement. However, do not try to “coax” them toward the lever by activating your shaper switch and providing reinforcement. If you do, you are actually reinforcing them for staying at the back of the Skinner box! Be patient. Don't let the rat take control of your behavior. There's always tomorrow and your rat will be thirstier then.

As a general rule, only the intermediate behaviors of approaching and touching the lever should be reinforced.

Being a good shaper: The method of training responses approximating the target behavior (*i.e., lever pressing*) is called *shaping by reinforcement of successive approximations*. Eventually, only the target response (*i.e., lever pressing*) will be reinforced with water, but we have to get to that point by a series of small steps. Recall that the process by which selected operant's are altered by the application of positive

reinforcers is called reinforcement. Can you verbalize the difference between the concepts of reinforcement and a positive reinforcer?)

The trick to being a good shaper of behavior is to consistently reinforce only the behavior you desire. Be patient and consistent.

Shaping is by far the hardest thing you will do in this lab, and as previously stated, its importance to the rest of these experiments and the importance of your role in the shaping process cannot be overstated. Have you ever taught a dog to sit on command? Not easy is it? Well, this is essentially what we're doing here, except that most dogs are smarter than your rat. Further, dogs sit all the time in the wild, but rats never push bars in the wild to get water. Given that, this is not going to be easy, but its level of difficulty is directly related to your performance.

To successfully shape by successive approximations, you must be very observant and methodical – reinforcing only those behaviors that are appropriate. If you fail to reinforce the right behavior at the right time, you will have a very difficult time shaping your rat. If this occurs and the other students in the lab are not having these same problems, you can be assured that the fault is yours and not because your rat is dumb. For the most part, the intelligence level of these rats is the same, which means that these animals are going to respond exactly the same to shaping. Therefore, the difference in time that it takes to shape these animals lies mostly in the knowledge and skill of the shapers.

Cutting the Apron Strings: A commonly asked question concerns when you should withhold reinforcement for a previously reinforced behavior (*i.e., you were reinforcing the rat for sniffing the bar, but now you want them to do more than just sniffing it*). When should you start withholding reinforcement for a previous (*close, but no cigar*) behavior and reinforcing only those behaviors that are closer to the target response (*lever pressing*)? It is a judgment call. If you think your rat's overall activity is lever oriented, just keep shaping by successive approximations and don't reinforce earlier response forms. You must be patient and very observant. The bottom line is that shaping is not an exact science. Successful shaping has occurred when your rat is reliably performing the target response of lever pressing.

What has the rat learned at this point?

- where the lever and water are located
- when water is available
- the response contingency – that is, which target response is associated with positive reinforcement

Experimental Activity and Data Collection: How do I quantify what I am doing? Shaping, like magazine training, is not particularly amenable to quantification or to plotting results. Further, without a lab partner present, it is difficult to record and shape at the same time. Luckily, with the new digital, computer-controlled equipment in the lab, all you must worry about is shaping your rat because the computer will tabulate your data for you! (*The lab session instructor will upload the data from the computer to Canvas*).

After the lab is over and with this data, you can determine the number of reinforcements and responses that occurred, thereby allowing you to calculate the ratio of shaping responses to reinforcements for each block of time. The closer to one (1) that that ratio is, the more reinforcements are the result of correct responses and the more precisely your rat has been operantly conditioned to lever pressing.

You may observe that as you progress through this session, more and more of the rat's behavior will be reinforced. You may also notice that the animal is learning what not to do (*i.e., all those behaviors that do*

not produce reinforcement). Sometimes, after the rats have been successively approximated to the vicinity of the lever, they will progress very rapidly (*that is, they will begin lever pressing without any further shaping required*). If this occurs, congratulations.

Additional Notes

- Shaping is an “art” and it is not easily quantified.
- Shaping may take several lab sessions to accomplish (*we have scheduled six, which is more than sufficient IF you and your partner do your part by being good shapers*).
- Don’t be frustrated; you must be patient and consistent and you must know when to reinforce and when not to. These are the key components in this process.
- If your neighbor’s rat is bar pressing within 10 minutes and yours hasn’t yet done so by the fourth day, it probably means that you were not a good shaper (*i.e., you failed to be quiet, observant, patient, and to reinforce and withhold reinforcement as appropriate*).
- It is essential that shaping is complete before you can begin experiment 4, so if you need to come in on Friday for an additional shaping session, schedule that with your lab instructor. Also, leave the water off your rat’s cage and make a note on the cage stating that you are coming in on Friday to conduct an additional shaping session. Doing this will help ensure that your rat doesn’t receive any water on Thursday afternoon.

Assignment 1: Handling, Magazine Training, and Shaping

Graph(s):

- Construct a behavioral trendline graph for this experiment (*you will turn in one graph that contains data over 4 days of training*). See Appendices for detailed instructions.

Short-answer questions:

- In Experiment 1, you were required to handle the rat. What is the goal of handling and how will the results of your experiment be affected if you had not handled the rat prior to the other experiments? Give specific examples.
- In Experiment 2, the goal of this exercise is to classically condition the rat to locate and successfully partake of the reinforcement (*water*). For magazine training, what is the US, CS, and UR?
- In Experiment 3, you were required to shape your rat’s behavior. Explain your methodology for reinforcing successive approximations of lever-pressing behavior, then moving toward non-reinforcement of earlier response forms. What worked and what did not, and why do you think various strategies worked or didn’t? Explain.
- How does magazine training help in shaping? Explain.

Experiment 4: Continuous Reinforcement (CRF)

The Skinner box and computer program allow you to manipulate both the rate and patterning of your rat's lever pressing responses, and when animals are subjected to different schedules of reinforcement, distinctive patterns of lever pressing result.

Continuous Reinforcement: The first schedule of reinforcement you will train your rat on is continuous reinforcement (CRF). In continuous reinforcement, each lever pressing response is reinforced with water. Note that CRF was also the end point of the previous lab. Note: As you might recall from your Intro to Psychology course, FR-X is a fixed ratio schedule of reinforcement, with ratio meaning the number of responses that are required for reinforcement and "X" indicating the number of responses that are required. Since CRF means that every response results in reinforcement, CRF is the same as FR-1.

The acquisition of this new pattern of lever pressing behavior proceeds in a predictable manner. Not unlike the acquisition of a Pavlovian conditioned response, CRF schedules characteristically produce positively accelerated patterns of response. A positively accelerated slope simply means that as time passes, increasingly more responses are made and less time is taken between responses. After training, lever pressing levels off at a rate dictated by how rapidly the animal can drink the positive reinforcer.

Experimental Activity and Data Collection: Once again, the boxes and the computer will do most of the work for you. From the data you obtain from Canvas, you will plot a cumulative record displaying your rat's lever-pressing performance in time. A graph of the cumulative record will (*hopefully!*) show you the positively accelerated slope mentioned above.

Our next experiment is extinction and spontaneous recovery. How do you think a rat will perform if it was conditioned with CRF but is now experiencing an extinction event? How would a CRF-conditioned rat perform in comparison to one that has been conditioned on an intermittent schedule of reinforcement? Let's answer those questions by using some examples. If you put the right amount of money into a soda machine and it doesn't give you a soda, what do you do? Do you just casually accept that outcome or do you pound on the machine a bit? Are you likely to put more money into the machine or are you more likely to walk away, confused, frustrated, and feeling cheated? On the other hand, what would happen if you didn't really expect to receive a soda every time that you put your money into the machine? What if your experience has been that it sometimes gives you a soda and sometimes not? Consider the gambler playing a slot machine. They put their money into the machine, pull the arm, the tumblers spin, and sometimes get lots of coins but most often the machine just eats their money. And yet, they play again. With partial (intermittent) reinforcement, the subject has lived with non-reinforcement for their behavior and they also never know when reinforcement is coming. With CRF (FR-1) conditioning, the subject EXPECTS reinforcement EVERY time they perform the required behavior. Thus, the first non-reinforced behavior is unexpected (shocking) and completely intolerable. But not so for an animal that has been trained on an intermittent schedule of reinforcement.

Experiment 5: Extinction and Recovery

If you haven't yet covered extinction and spontaneous recovery in lecture, read the relevant material in your course textbook, or if you are not enrolled in the lecture class, in some other source (*e.g., a textbook, internet*). You'll be glad you did!

This laboratory exercise is divided into three parts.

- Part 1: During the first ten minutes, your rat will be responding on CRF schedule, just as in the previous experiment.
- Part 2: During the next ten minutes, reinforcement will be inactivated. At this point, the lever pressing response will no longer be reinforced, a procedure called *experimental extinction*. Note that if your rat stops responding before the first ten minutes are up and doesn't respond again during the entire second ten minute period, true extinction hasn't necessarily occurred (*it might be due to premature satiation, fatigue, or any number of other factors*). Also, you might see spontaneous recovery occur during this 15-minute period.
- Part 3: Finally, in the last ten minute period, CRF will be reinstated and continued for the first ten lever presses, and then Fixed Ratio for 2 responses (*FR-2*) is programmed for the remainder of this period.

Assignment 2: Continuous Reinforcement, Extinction and Recovery

Graph(s):

- Construct a cumulative record graph each for Experiment 4 and Experiment 5 (*you will have ONE graph for each experiment that includes both your data and your partner's*).

Short-answer questions:

- In general, what does a cumulative record tell us about the rat's behavior during a training session? Give specific examples using various characteristics of the graph (*e.g. gradient*) and explain.
- In experiment 5, how quickly was your rat's lever-pressing behavior extinguished in part 2? That is, how many lever-presses did they perform during extinction before they ceased lever-pressing?
- What does this result tell you about the strength of your rat's operant conditioning to lever-pressing? Explain.

Experiment 6: Partial Reinforcement

Recall that you ended the previous lab session on FR-2, which is a partial schedule of reinforcement. If you had continued with CRF training, your animal would have developed strong expectations regarding the response/reinforcement contingency. However, during experiment 5, we introduced an extinction session and ended the session on FR-2, which violates these reinforcement expectancies. That is, in experiment 5, every response was not reinforced. Yes, your rat would be responding more efficiently, and at a faster rate, if we had kept it on CRF, and more time is now required to achieve steady lever pressing because of this previous extinction session. But as a result of that early extinction exposure, your rat is now better prepared to experience partial schedules of reinforcement, which along with exploring the partial reinforcement extinction effect (*PREE*), is the goal of this lab session.

You will also observe and quantify your rat's behavior during the last day of this lab experiment.

Effects of Partial Reinforcement: For present purposes, partial reinforcement schedules:

- result in a slower acquisition of stable responding;
- produce more responses in a session, presumably because the animal's thirst is slaked more slowly than on continuous reinforcement; and
- produces greater resistance to extinction (*this is called the partial reinforcement extinction effect, PREE*).

Training animals on partial reinforcement schedules: In this lab session, you and your partner will be training your rat on a partial reinforcement fixed ratio schedule. More specifically, the computer program will allow several reinforcements at FR-2 and then it will increase to FR-5, FR-8, and hopefully your rat will reach FR-38 by the end of this experiment.

The problem you might encounter when training animals on partial schedules of reinforcement is maintaining an appropriate density of reinforcement per unit time. As you will read in your course textbook (*or some other source if you are not enrolled in the lecture*), all animals are exquisitely sensitive in judging optimal work-to-payoff ratios. In the same way that you would be insulted if someone offered you \$1 per hour to scrub floors, rats will not typically press the lever hundreds of times for one little squirt of water.

Some Terminology: A CRF schedule is said to be *reinforcement dense* compared with a FR-38 schedule, which is considered *reinforcement lean*. You will be “leaning out” the schedule of reinforcement, which means moving gradually from a very dense schedule to a leaner schedule of reinforcement. This could be very challenging, so it is best accomplished in small stages. In fact, if you were to switch immediately from FR-2 to FR-38, extinction would likely occur. Luckily for you though, we have programmed the computer to automatically conduct smaller steps so that partial reinforcement occurs rather than extinction.

RULE 1: If the schedule gets too lean too quickly, your rat may decide to quit responding, which is known as *ratio strain*.

RULE 2: The longer your rat is on a particular schedule of partial reinforcement, the greater the animal's expectancies that the schedule will not change (*and the more difficult it is to switch the rat's behavior to a leaner schedule*).

Experiment 7: Partial Reinforcement Extinction Effect

As mentioned briefly in the previous experiment, the differential effects of partial reinforcement versus continuous reinforcement are many, but a notable benefit is that of a *greater resistance to extinction*.

Your textbook discusses this phenomenon with some detail, but uses slightly different terminology. On page 301, you will find that Powell, Symbaluk, and Honey (2009) begin a discussion on the *resistance to extinction*, and on page 302 refer to this phenomenon in bold and italicized font as ***the partial reinforcement effect***. Simply stated, “behavior that has been maintained on an intermittent (partial) schedule of reinforcement will extinguish more slowly than behavior that has been maintained on a continuous schedule.” (Powell et al., 2009).

While we unfortunately do not have the time in this class to demonstrate, research has also shown that not only would an animal trained on a PRF schedule resist extinction in a greater manner than one trained on a CRF schedule, but an animal trained on say, an FR100 would have an even greater resistance to extinction than an animal trained on an FR10.

G.S. Reynolds (1975) demonstrated that of all forms of partial reinforcement, subjects trained on a *variable ratio schedule* would have the greatest resistance to extinction. For example, an animal trained on an FR15 would have quicker extinction than one trained on a VR15 schedule. Due to the confines of our laboratory, especially regarding time, we unfortunately are unable to replicate Reynolds 1975 study, but we will spend two days devoting time to proving that the phenomenon known as the partial reinforcement extinction effect (*PREE*) does reliably exist.

In today’s experiment, we hope to exemplify the fact that PRF has a greater resistance to extinction than CRF by running a similar protocol to the one we used for Experiment 5 (*extinction and recovery to CRF training*) and comparing the results of your animal’s performance today after having been successfully training on a PRF schedule to the results of Experiment 5 (*after successful training on only a CRF schedule of reinforcement*). The comparison should show a more continuous pattern of pressing behavior from your animal.

Like Experiment 5, today’s protocol is divided into three phases. In the first 10 minutes of the experiment (*phase 1*), your animal will be pressing on an FR2 schedule. In Phase 2 of the experiment, we will enter experimental extinction for 10 minutes and your animal will be not be reinforced for any number of lever presses that are emitted. In phase 3, we will re-enter the FR2 schedule of reinforcement for the remaining 10 minutes.

Assignment 3: Partial Reinforcement and Partial Reinforcement Extinction Effect

Graph(s):

- Construct a cumulative record graph for the results of Experiment 6 and Experiment 7. This means you will construct TWO graphs, one for Experiment 6 (*with four days' worth of data*) and one for Experiment 7 (*with four days' worth of data*).
- Include a copy of your animal's graph from Experiments 4 & 5 when answering the 1st & 2nd short answer question below, respectively. Do not expect your instructor to dig up the assignment you turned in before to grade your comparisons.

Short-answer questions:

- Compare and contrast the results you saw between Experiment 4 and Experiment 6. How did your animal respond differently to a partial reinforcement schedule than a continuously reinforced schedule? Explain.
- Compare and contrast the results from Experiments 5 and Experiment 7. Describe the differences and similarities that are visible on your graphs and give explanations for those differences. Explain.
- (*Note, it may be worthwhile to place an asterisk or some other notation at each of these notable areas of differentiation so that the grader of your assignment can tell at which point in the experiments you are observing*).

Experiment 8: Choice Behavior

You can find ample discussion over the phenomenon of behavior when given a choice in your textbooks starting with page 380 and following. As your lecture text defines, the term *concurrent schedules of reinforcement* consist “of the simultaneous presentation of two or more independent schedules, each leading to a reinforcer.” (Powell et al., 2009). What this means for our lab in this experiment is that your animal can be offered the opportunity to press two levers each with reinforcement schedules independent of one another.

This is exactly what we intend to do. Before we do so, however, we must introduce the concept of a lever on the right side of the feeder trough to the animal. Remember that your animal has become accustomed to pressing a lever that is located to the left side of the trough only. To simply add a lever on the right side might result in data that is not completely reliable to true choice behavior since we cannot be sure that the acquisition of “left lever pressing behavior” can generalize immediately to “right lever pressing” as well. For this reason, we must introduce a lever on the right side of the trough and remove the lever from the left side. During this **choice priming event**, we will place the right lever on a FR2 schedule of reinforcement to help establish the behavior of pressing a lever on the right side of the trough.

After four days of priming with the right lever only, we will reintroduce the lever that was originally on the left side, leaving the right lever in its current place. Now the animal will have the option of choosing which lever to press. The next question to ask ourselves is, “What would the animal do if each lever were on different schedules of reinforcement?”

Richard Herrnstein and colleagues (1961) discovered that animals presented with two different reinforcement possibilities will generally match their rate of responding with the schedules presented to them in order to receive the maximum benefit. Regardless of the reinforcement schedules presented on each lever, a fully trained animal will essentially go back and forth between the levers pressing in a manner that will grant maximal reward. If maximum reward comes from only pressing one lever, then a fully trained animal would totally disregard the opposing lever. This might only occur in the event that both levers are on a FR schedule and one lever has a much leaner schedule than the other.

This effect has subsequently become known as **Herrnstein’s Matching Law**. Your lecture textbook refers to this only as *the matching law*.

Behavioral matching can be accomplished with multiple combinations and mixtures of schedules, that is, when given enough training. For instance, the animal would have to choose how to distribute its behavior between a variable ratio and a fixed interval schedule. This is what we hope to examine in this week’s experiment.

In this experiment, your rat will have the opportunity to investigate two different reinforcement schedules. More specifically, we will use FR and FI schedules in this experiment, and if there was sufficient time, we would allow this experiment to continue for more than just four days and also perhaps manipulate the densities in order to more thoroughly test Herrnstein’s law.

If your rat is not performing, you and or your partner need to let your instructor know as they might be able to manipulate the densities of these schedules of reinforcement in order to get better results. In an actual research lab, such manipulation would likely not occur because the intent is often to investigate the

amount of time that is required for matching to occur. However, in this lab, we want you to see this phenomenon occur, so we might need to adjust the reinforcement densities.

With all of this said, we will first start with the left lever on a FI-30 (*seconds*) and the right lever on a FR-15. However, if the animals seem to be having a hard time by their second day of training, then we might increase these densities or the difference between them.

Assignment 4: Choice Behavior

Graph(s):

- Construct a behavioral trendline for Experiment 8 week 2 (*you will have ONE graph for the experiment that includes both your data and your partner's – thus four days' data in one graph*).

Short-answer questions:

- What is the purpose of the priming event? Was it efficacious in your experiment? What do you suppose would have happened had we not employed the choice priming event? Would your animal have performed in a manner that guaranteed maximal reward? Explain.
- Describe your data in this experiment. Describe how your observations throughout this experiment compared to your predictions. Were you right in your hypothetical estimations? If you had the opportunity, what might you have done differently to best guarantee maximal reinforcement? Explain.

Experiment 9: Stimulus Discrimination

One goal of behaviorism is to identify and analyze the way in which environmental stimuli exert control over behavior. By requiring lever pressing in a distinctive environment through selective positive reinforcement, you exerted stimulus control over your rat and this can potentially be used to make highly accurate, reliable predictions that a particular response will occur in a particular situation.

In the next few sessions you will further demonstrate stimulus control by running S+/S- discrimination trials. Recall that S+ is an abbreviation for a positive contingency between a behavior and reinforcement (*i.e., if the behavior is performed, reinforcement follows*), while S- indicates a negative contingency between a behavior and reinforcement (*i.e., the behavior will not be reinforced*). If you haven't covered this in lecture yet, it would be a good idea to read about this in your course textbook or in some other source (*e.g., internet*).

Stimuli in Skinner boxes: Placing a trained, water-deprived animal into the Skinner box provides the occasion for specific operants to occur. That is, the sights, sounds, and smells of the box are environmental stimuli (*i.e., the context*) that have helped shape the lever pressing behavior. Therefore, your previous lab sessions have already begun the process of training stimulus discrimination. That is, in the previous experiments, the lighted room, Skinner box, lever, ambient lights, and background noises collectively set the occasion for lever pressing and subsequent reinforcement. In other words, your rat has already learned that lever pressing in this environment produces reinforcement (S+) and that reinforcement does not occur outside of these environmental influences. Our task now is to get the rat to recognize and distinguish between S+ and S- influences within this environment. More specifically, we will use the white house light to indicate that reinforcement is available for lever pressing (S+), and we will put the animal into the dark by extinguishing these lights when reinforcement is not available (S-). Stimulus control has been accomplished when your rat responds to the S+ condition and does not respond to the S- condition.

An organism's behavior is said to be under stimulus control when it responds one way in the presence of a stimulus, and another way in the absence of that stimulus.

S+/S- Trials: This experiment begins by reinforcing lever pressing responses during an initial five minute S+ period on a dense schedule of reinforcement (*in this experiment, CRF*). This is then followed by a five minute S- condition, and these different conditions then alternate for each five minute block of time. Therefore, the programming for your 30-minute training session is scheduled like this:

Time (min)	S+ or S- Condition?	House Lights	Operant Conditioning chamber is dark?	Reinforcement is scheduled to occur?
0 – 5	S+	On	No	Yes
5 – 10	S-	Off	Yes	No
10 – 15	S+	On	No	Yes
15 – 20	S-	Off	Yes	No
20 – 25	S+	On	No	Yes
25 – 30	S-	Off	Yes	No

These thirty-minute training sessions will be continued for several days (*see calendar of events*). At some point, you rat will (*hopefully*) respond to the S+ condition but not to the S- condition.

A Couple of Relevant Questions:

Question: How many sessions will it take my rat to learn this discrimination?

Answer: That depends upon you, your rat, and its past reinforcement history. On this question, the best answer is often "several."

Question: If, after my rat learns the S⁺/S⁻ discrimination, the conditions were reversed such that it now gets reinforced for bar pressing when the lights aren't flashing (and not reinforced for bar pressing when they are flashing), would my rat learn this reversal?

Answer: Yes, Harry Harlow did a whole series of these discrimination reversal problems, and yes, even rats can learn to reverse discriminations. In fact, monkeys can learn to reverse discriminations so well that they eventually will reverse on the basis of one or two reinforced (or non-reinforced) responses. But rats never get that good.

Assignment 5: Stimulus Discrimination

Graph(s):

- Construct a cumulative record graph that includes both your data and your partner's).

Short-answer question:

- On which day did your rat seem to adjust its behavior? In other words, when did it respond according to the required stimulus discrimination? Explain.

Rat Olympics

In this project you will work with your partner to teach your rat a trick of your choice. You and your partner should choose a trick that is interesting, yet simple enough to be accomplished in a few days.

Protocol: The rats will be water deprived as usual. Water reinforcement via a pipette will be provided for the training of your rat.

Grading: Your Rat Olympic (RO) grade is not affected by your rat's performance. Rather, your RO grade is determined by the quality of your effort in designing and executing the target behavior (*as indicated by your RO preliminary report*) and by your assessment of what went right or wrong with your rat's performance of the target behavior (*as indicated by your RO final report*).

Examples:

Teaching your rat how to spin on the spot

<http://www.youtube.com/watch?v=iljHxtk7XZI>

Teaching your rat how to walk on its hind legs

<http://www.youtube.com/watch?v=nNOgpQLDZH8&feature=related>

These are merely suggestions. The possibilities of what you can achieve are endless and bound only by your imagination. In the past, rats have played basketball, bowled, walked a tightrope, climbed a rope, etc. Be Creative!!



Reinforcers: You are welcome to choose a reinforcer of your choice as long as it is rat friendly. Most students use either cereal or sugar water as the reinforcer. If you choose to use sugar water, the lab will provide it and pipettes to administer it. Reinforcers that aren't allowed are things such as candy or peanut butter. If you are unsure if your selected reinforcer is rat safe please read what [are unsafe foods for rats](#). You can also ask your TA if it is safe to use and it is up to the TA to approve it.

Containers: You and your partner are responsible for bringing in a container to train your rat in for Rat Olympics. This container must be large enough that your rat cannot jump out of it and have enough space for you to be able teach your rat your chosen trick. We suggest a 28-to-20-gallon plastic tote. Dimensions of your container should not be lower than 23 1/8" x 18 1/8" x 14 7/8". If you are unsure of whether your container meets the requirements for Rat Olympics please check with your TA. Additionally, all containers must be brought into lab by either you or your partner by the specified date on the calendar. Failure to bring in your tote by that date will result in a 5-point deduction on your Rat Olympics Final report.



Rat Olympic Reports

Rat Olympic Prospectus:

- The purpose of this written submission is to ensure that your project is doable, and it should consist of: A bullet point format of your proposed idea for the Rat Olympics. You and your partner may turn in the same report, but the parameters of your task must be clearly described.

Preliminary Write-Up: Your written description should include:

- A detailed description of your objective, well written and in paragraph form.
- A detailed description of methods, apparatus, and the specific reinforcement(s) that you will use to achieve the objective.
- A log of your training sessions (how many hours the animal required to learn its task).

Rat Olympic Final Report: Your written final report should include:

- A detailed description of the results of the experiment.
- An analysis of any confounding variables, if applicable (*why it didn't work*).
- How you were able to successfully achieve your objective.
- Your final report must be clear and comprehensive.
- In order to achieve these requirements, your write-up should be a minimum of two (2) pages, single spaced.
- Your report should follow the formatting described in your syllabus. Failure to meet formatting expectations will result in deduction of points.

See the calendar for the due dates for your Rat Olympic Preliminary Write-Up and Final Report.

Appendix 1.0: Basic Graphing Instructions and Walk-Through

These instructions will aid you in completing the graphs for each of your other assignments. For your convenience, instructions are included for both the Windows and Mac operating systems. Specific instructions for making a Behavioral Trendline and Cumulative Record graph are included in additional appendices. If you encounter problems, please seek help from your lab instructor in a timely manner.

Basic Graphing Instructions

- For every graph, be sure to label all axes and include a graph title.
- Identify each point/line using a legend.
- Submit graphs using different colors, legends, and titles as this will best differentiate your graph. See your lab instructor for specifics on what they prefer for you to do in the case color is not an option.
- If you have any difficulty creating a graph, it is **YOUR RESPONSIBILITY** to contact your instructor for help. Ignorance is not a valid excuse for poorly constructed or erroneous graphs.

Appendix 2.0: Instructions for Graphing Your sniffy Results with Excel on Windows

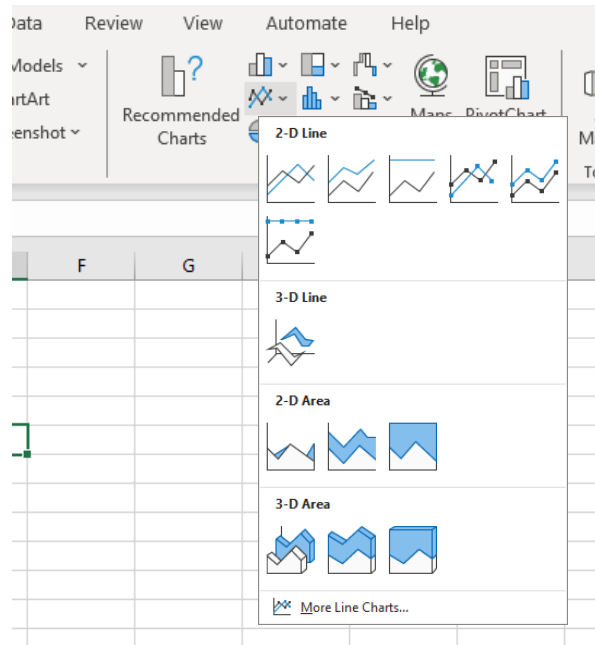
1. First, read your Sniffy section of this lab manual about using a spreadsheet program like Excel to graph your results. After completing the Sniffy Lab Manual instructions about opening an Excel Spreadsheet, it is time to create a graph. Your spreadsheet should look like this:

A	B	C	D
Exercise 4			
Trial	Weak CS	Medium CS	Strong CS
1	0.144	0.3	0.504
2	0.151	0.5055	0.9576
3	0.1196	0.6463	1.2902
4	0.0848	0.7427	1.5147
5	0.0567	0.8087	1.6603
6	0.0367	0.854	1.7526
7	0.0232	0.885	1.8105
8	0.0144	0.9062	1.8464
9	0.0089	0.9208	1.8688
10	0.0054	0.9307	1.8825

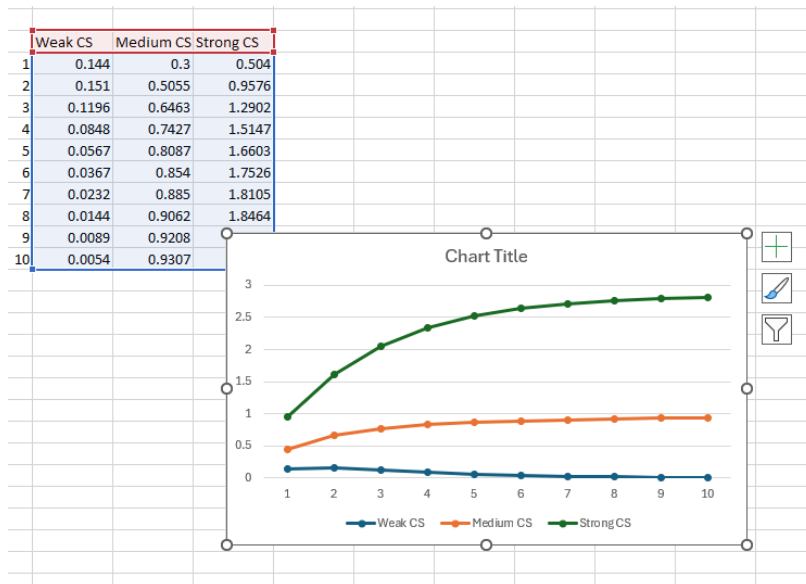
2. The most crucial step in making the graph is to highlight the correct data. Highlight all the values at once, along with the titles.

A	B	C	D	E
Exercise 4				
Trial	Weak CS	Medium CS	Strong CS	
1	0.144	0.3	0.504	
2	0.151	0.5055	0.9576	
3	0.1196	0.6463	1.2902	
4	0.0848	0.7427	1.5147	
5	0.0567	0.8087	1.6603	
6	0.0367	0.854	1.7526	
7	0.0232	0.885	1.8105	
8	0.0144	0.9062	1.8464	
9	0.0089	0.9208	1.8688	
10	0.0054	0.9307	1.8825	

3. Now it is time to use the line graph generator that Excel provides to easily make this into a graph.
 - Click on the tab labeled “Insert”. In the section labeled “Charts” you will find a display of different graphing options.
 - Since the Sniffy Manual displays a line graph, click this option.
 - As you can there are many kinds of graphing options, the stacked line option is ideal for this purpose.



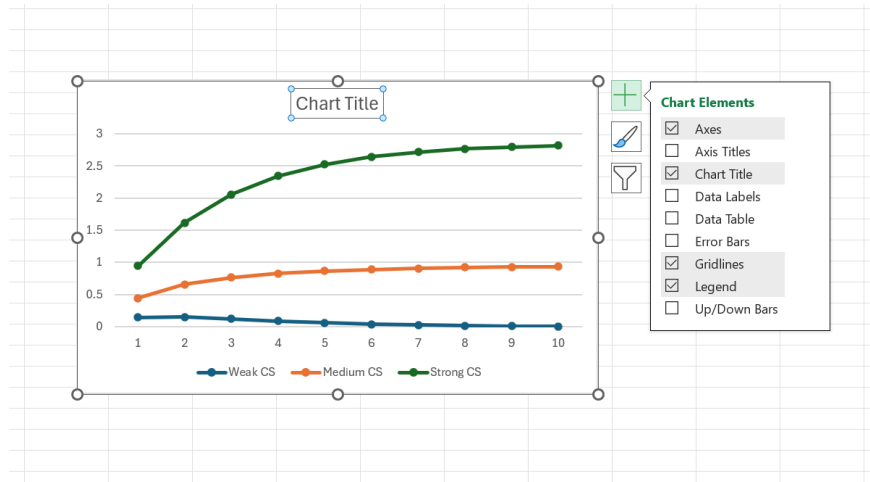
4. Clicking on the Stacked Lines with Markers option, should generate the graph automatically.



As you can see, excel automatically added the labels to the lines.

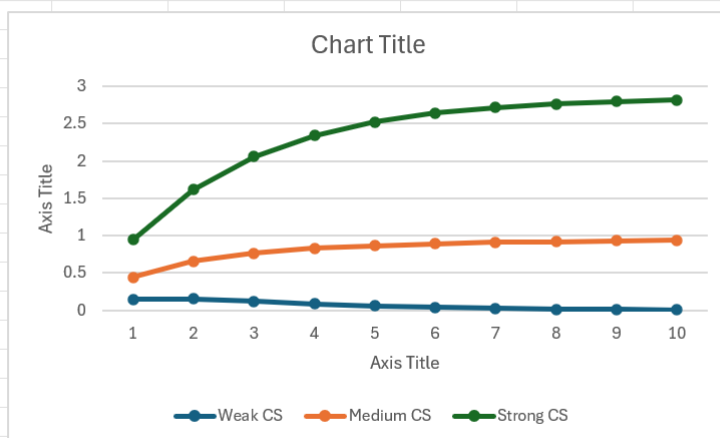
5. Now it is time to Format

- To add Axis labels click on the graph and then select the plus sign to see the “Chart Elements

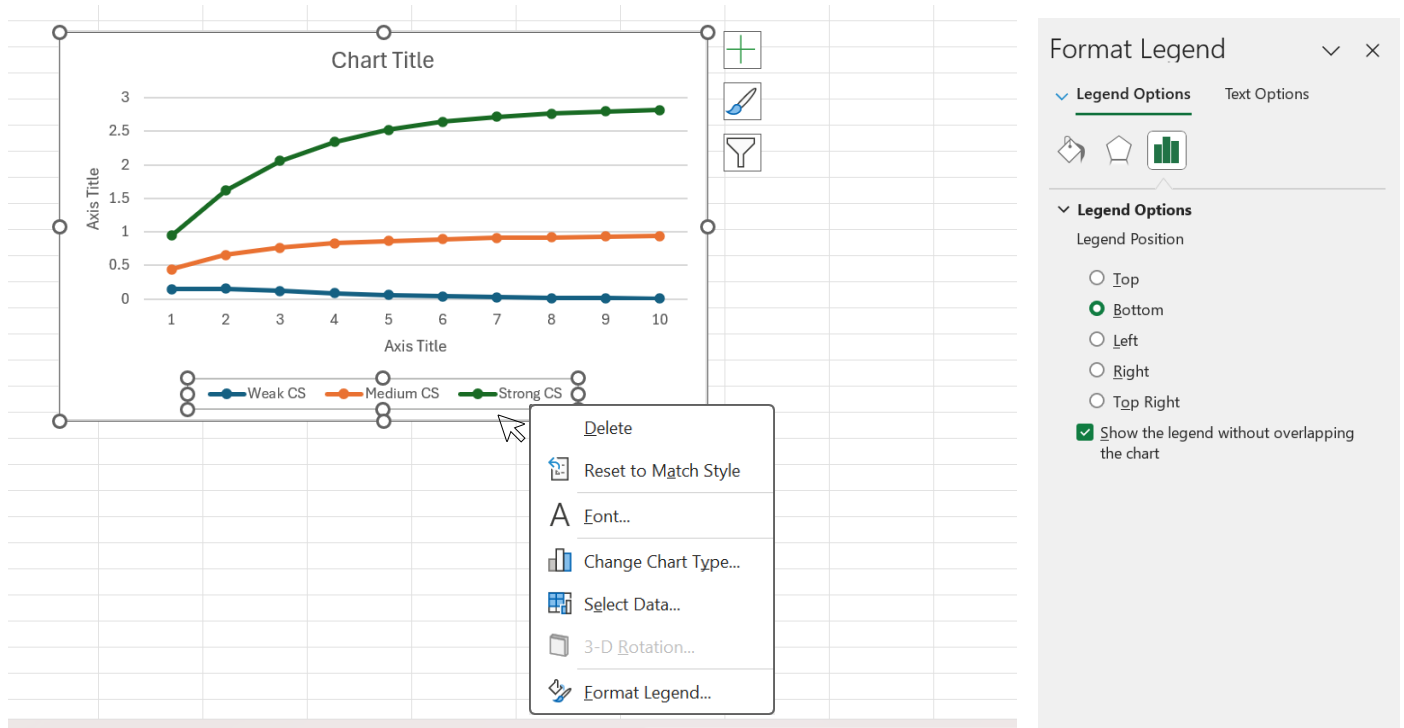


- Make sure there is a checkmark on the following:
 - Axes
 - Axis Titles
 - Chart Title
 - Gridlines
 - Legend

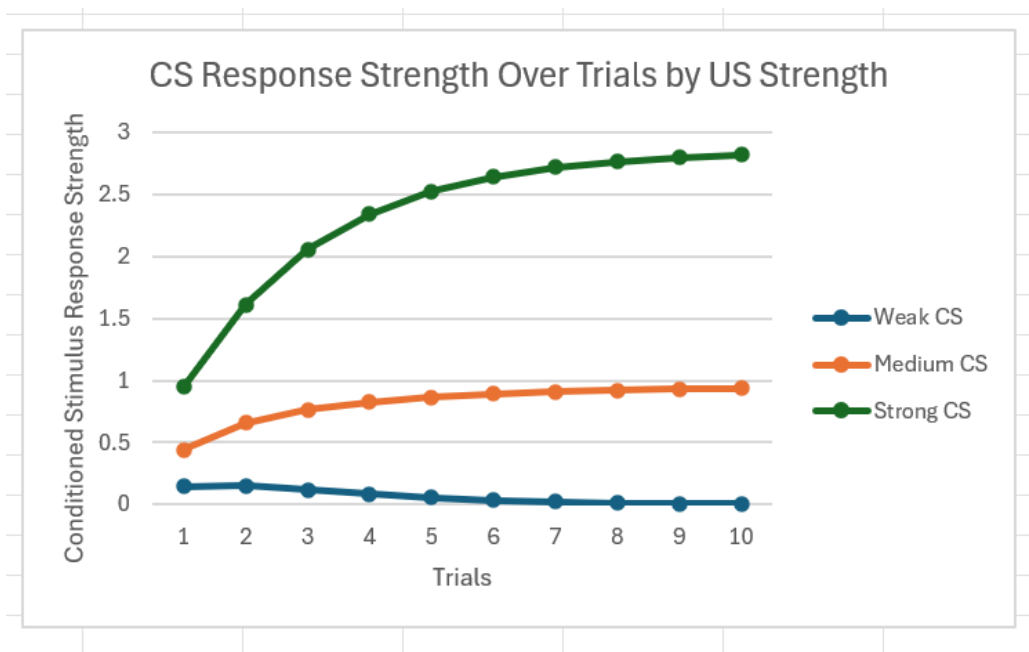
Your graph should now look like the following:



6. To move the legend to the right of the chart for better formatting, right click on the legend text box and select “Format Legend”. This should bring up a menu on the right side of the screen where you can choose where you want to have the legend.



Example Final Graph:



Appendix 3.0: Instructions for Data Sorting

Data Clean Up:

- To open the text file, first locate the correct file under the “Files” section on Canvas. Right-click and select “download linked file” or “download linked file as” to save the file to your computer. Open on Excel to begin the assignment.
- For all experiments throughout the semester, you may need different data (therefore different columns) depending on the type of graph needed for each experiment. However, examples of the data rows are shown below. Either copy the data for your specific rat (subject number) to a different spreadsheet, or delete the remaining columns (the ones that do not pertain to your subject number) that you will not need.
 - Subject
 - Box
 - Start date
 - Start time
 - End time
 - Session Duration
 - Total rewards Earned
 - Rewards via L. Lever
 - Rewards via R. Lever
 - Rewards via Shaper Switch
- Depending on the experiment, you may further delete other unnecessary columns (see detailed instructions for each graph type below).
- The next step is to isolate the data from your animal. The data is organized by subject number and the date of the experiment so search for your designated subject number to find your data. You may do this either by using the Find command (Ctrl+F/Cmd+F) or by sliding the page over. After you have located your animal’s data, highlight and copy all the relevant information and columns to a new spreadsheet or delete the data that does not pertain to you.

Your clean data should look something like this:

	A	B	C	D	E
1	Subject	1	1	1	1
2	Box	1	1	1	1
3	StartDate	#####	#####	#####	#####
4	StartTime	7:54:31	8:00:04	7:46:39	7:58:19
5	EndTime	8:33:36	8:45:50	8:30:28	8:36:52
6	Session Duration	1800.01	1800.01	1800.01	1800.01
7	Total Rewards Ea	31	105	157	191
8	Rewards via L. Le	1	34	151	188
9	Rewards via Shap	30	71	6	3

- Now that you’ve filtered out the basic data, see below for the instructions to whichever particular type of graph you are going to create.

Appendix 3.1: Behavioral Trendline Graph

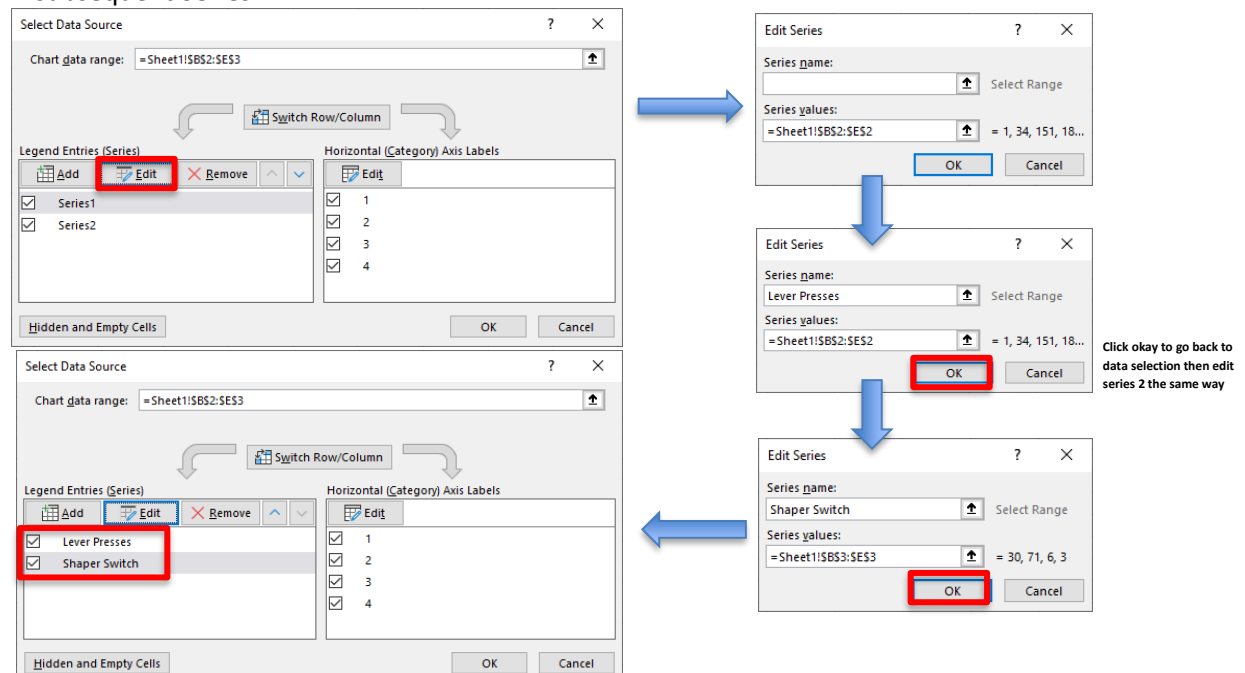
- The behavioral trendline graph allows you to look at the totals for each particular behavior your animal partakes in over the course of several days. For example, during shaping, you are comparing the total number of lever presses to the number of shaper switch presses. The example of shaping will be used throughout this walk through.
- Clean up the data as instructed in Appendix 3.0 leaving only the columns corresponding with the behavior to be graphed. In the example of our shaping experiment, you will eliminate all rows except the following:
- Original Row: 1 – Subject
- Original Row 8 – Rewards via L. Lever
- Original Row: 9 – Rewards via Shaper Switch

Your clean data should look something like this:

	A	B	C	D	E
1	Subject	1	1	1	1
2	Rewards via L. Lever	1	34	151	188
3	Rewards via Shaper Switch	30	71	6	3
4					

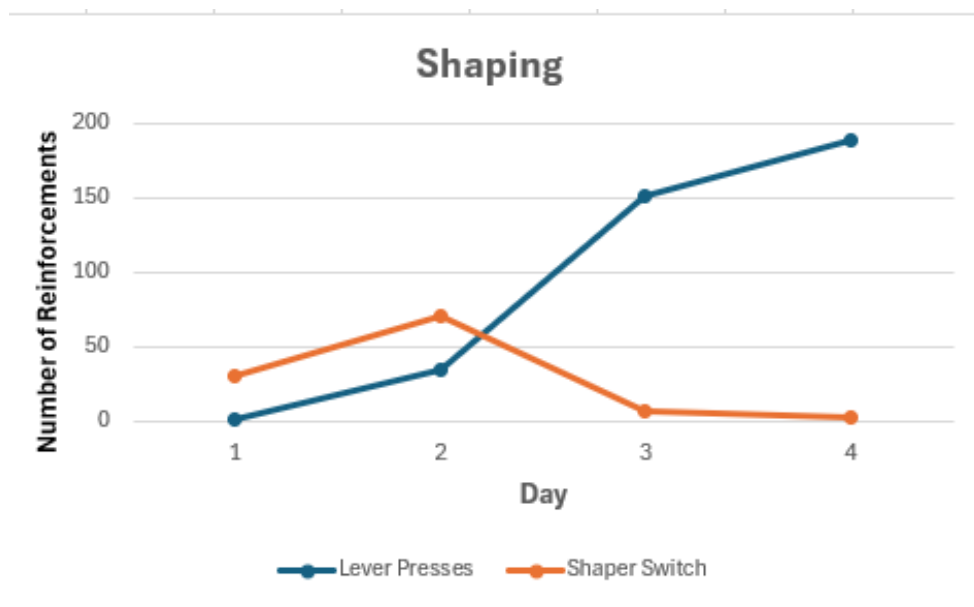
Remember, in other experiments requiring a behavioral trendline graph, you may be choosing a different selection of behaviors.

- To graph your data, highlight the values for lever and switch (exclude headings), click on the “insert” tab at the top of the page, in the area labeled "Charts". Since the Sniffy Manual displays a line graph, choose the line chart.
- Once your line chart has appeared on the page press on the plus sign  to see the Chart Elements.
- Make sure to select “Chart Title” and “Axis Titles.” Label your Title and Axis accordingly.
- To edit the graph legend (now labeled “Series”), right click on the graph and choose “Select Data...”.
- Select the first Series 1, choose “Edit”, and rename appropriately. Do the same with each subsequent Series.



Click okay to go back to data selection then edit series 2 the same way

- Make stylistic changes as directed.
- Your final product should look something like this (*with different data points of course*).



Appendix 3.2: Cumulative Record Graph

- Cumulative Record Graphs only require you to graph lever presses as a function of time. Clean up the data as instructed in Appendix 3.0, leaving only the columns for your specific rat:
 - Column A – Data Labels (Subject, Box, Start Date, etc.)
 - Column B -- Left Lever Presses Time Stamps (*or Right Lever, if applicable*)

	A	B	C
1	Subject	1	
2	Box	1	
3	StartDate	9/23/2024	
4	StartTime	7:43:36	
5	EndTime	8:28:59	
6	Session Duration (sec.)	1800.01	
7	Total Rewards Eaned	228	
8	Total L Lever Presses	356	
9	L Lever Presses Time Stamps		
10	C(0)	21.17	
11	C(1)	37.58	
12	C(2)	47.16	
13	C(3)	49.84	
14	C(4)	50.15	
15	C(5)	50.26	
16	C(6)	55.33	
17	C(7)	58.11	
18	C(8)	59.61	
19	C(9)	60.11	
20	C(10)	60.86	
21	C(11)	63.44	
22	C(12)	74.15	
23	C(13)	84.09	
24	C(14)	84.77	
25	C(15)	84.93	
26	C(16)	96.01	
27	C(17)	96.29	
28	C(18)	99.68	
29	C(19)	102.89	
30	C(20)	108.15	
31	C(21)	108.35	
32	C(22)	109.47	
33	C(23)	112.71	

- The next step is to remove all the data included that does not pertain to a lever press (any cells in column B that do not include a time stamp). To do so, scroll down to the last time stamp number in column B, highlight the C(x) cells column and delete them.

350	C(340)	1633.09	
351	C(341)	1635.96	
352	C(342)	1636.47	
353	C(343)	1641.24	
354	C(344)	1641.8	
355	C(345)	1644.21	
356	C(346)	1645.27	
357	C(347)	1648.24	
358	C(348)	1649.16	
359	C(349)	1649.45	
360	C(350)	1719.84	
361	C(351)	1722.43	
362	C(352)	1729.71	
363	C(353)	1732.44	
364	C(354)	1734.8	
365	C(355)	1735.57	
366	C(356)		C(356)
367	C(357)		C(357)
368	C(358)		C(358)
369	C(359)		C(359)
370	C(360)		C(360)
371	C(361)		C(361)
372	C(362)		C(362)
373	C(363)		C(363)
374	C(364)		C(364)
375	C(365)		C(365)
376	C(366)		C(366)
377	C(367)		C(367)
378	C(368)		C(368)
379	C(369)		C(369)
380	C(370)		C(370)
381	C(371)		C(371)
382	C(372)		C(372)
383	C(373)		C(373)
384	C(374)		C(374)
385	C(375)		C(375)
386	C(376)		C(376)
387	C(377)		C(377)

- To make sense of the Time column, we first need to convert the remaining cells with “C(x)” into cumulative lever presses. To do this, create a new column and label it “Lever Presses”, begin typing

“1”, “2”, and “3” in the cells parallel to the C(x) rows. Then highlight these and drag down until the end of the “C(x)” rows.

	A	B	C	D	E
1	Subject	1			
2	Box	1			
3	StartDate	9/23/2024			
4	StartTime	7:43:36			
5	EndTime	8:28:59			
6	Session Duration (sec.)	1800.01			
7	Total Rewards Eaned	228			
8	Total L. Lever Presses	356			
9	L. Lever Presses Time Stamps			Lever Presses	
10	C(0)	21.17		1	
11	C(1)	37.58		2	
12	C(2)	47.16		3	
13	C(3)	49.84			
14	C(4)	50.15			
15	C(5)	50.26			
16	C(6)	55.33			
17	C(7)	58.11			
18	C(8)	59.61			
19	C(9)	60.11			
20	C(10)	60.86			
21	C(11)	63.44			
22	C(12)	74.15			
23	C(13)	84.09			

	A	B	C	D	E	F
339	C(328)	1576.09		329		
339	C(329)	1576.96		330		
340	C(330)	1585.23		331		
341	C(331)	1587.83		332		
342	C(332)	1588.97		333		
343	C(333)	1592.67		334		
344	C(334)	1595.64		335		
345	C(335)	1597.98		336		
346	C(336)	1601.25		337		
347	C(337)	1602.94		338		
348	C(338)	1609.25		339		
349	C(339)	1629.65		340		
350	C(340)	1633.69		341		
351	C(341)	1635.96		342		
352	C(342)	1636.47		343		
353	C(343)	1641.24		344		
354	C(344)	1641.8		345		
355	C(345)	1644.21		346		
356	C(346)	1645.27		347		
357	C(347)	1648.24		348		
358	C(348)	1649.16		349		
359	C(349)	1649.45		350		
360	C(350)	1719.84		351		
361	C(351)	1722.43		352		
362	C(352)	1729.71		353		
363	C(353)	1732.44		354		
364	C(354)	1734.8		355		
365	C(355)	1735.57		356		
366						
367						
368						

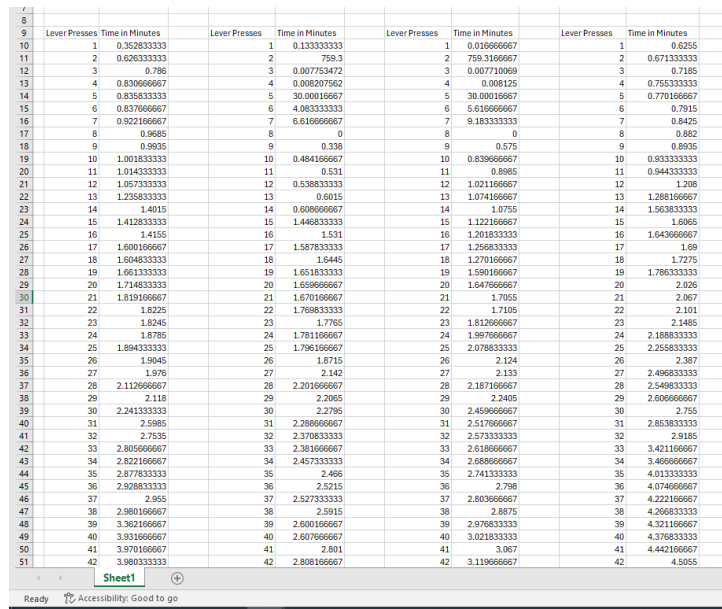
- Next, we must convert the time stamps in seconds given into minutes. To do this, create a formula dividing the time column (B2) by 60. In cells next to your cumulative presses, press “=”, select cell B10, then type “/” and “60” Your formula should read “=B10/60”. Press Enter. Label this new column “Time in Minutes.”
- When the cell is selected, you will notice that a green box appears when the mouse is placed over the bottom right-hand corner of the cell. Either double-click or drag this down the column to fill the remaining cells in the column with the formula.

	A	B	C	D	E	F
1	Subject	1				
2	Box	1				
3	StartDate	9/23/2024				
4	StartTime	7:43:36				
5	EndTime	8:28:59				
6	Session Duration (sec.)	1800.01				
7	Total Rewards Eaned	228				
8	Total L. Lever Presses	356				
9	L. Lever Presses Time Stamps			Lever Presses	Time in Minutes	
10	C(0)	21.17			=B10/60	
11	C(1)	37.58				
12	C(2)	47.16				
13	C(3)	49.84				
14	C(4)	50.15				
15	C(5)	50.26				
16	C(6)	55.33				
17	C(7)	58.11				
18	C(8)	59.61				
19	C(9)	60.11				
20	C(10)	60.86				
21	C(11)	63.44				
22	C(12)	74.15				
23	C(13)	84.09				
24	C(14)	84.77				
25	C(15)	84.93				
26	C(16)	96.01				
27	C(17)	96.29				
28	C(18)	99.68				
29	C(19)	102.89				
30	C(20)	109.15				
31	C(21)	109.35				
32	C(22)	109.47				
33	C(23)	112.71				
34	C(24)	113.66				
35	C(25)	114.27				
36	C(26)	118.56				
37	C(27)	126.76				
38	C(28)	127.08				

	A	B	C	D	E	F	G
1	Subject	1					
2	Box	1					
3	StartDate	9/23/2024					
4	StartTime	7:43:36					
5	EndTime	8:28:59					
6	Session Duration (sec.)	1800.01					
7	Total Rewards Eaned	228					
8	Total L. Lever Presses	356					
9	L. Lever Presses Time Stamps			Lever Presses	Time in Minutes		
10	C(0)	21.17		1	0.352833333		
11	C(1)	37.58		2			
12	C(2)	47.16		3			
13	C(3)	49.84		4			
14	C(4)	50.15		5			
15	C(5)	50.26		6			
16	C(6)	55.33		7			
17	C(7)	58.11		8			
18	C(8)	59.61		9			
19	C(9)	60.11		10			
20	C(10)	60.86		11			
21	C(11)	63.44		12			
22	C(12)	74.15		13			
23	C(13)	84.09		14			
24	C(14)	84.77		15			
25	C(15)	84.93		16			
26	C(16)	96.01		17			
27	C(17)	96.29		18			
28	C(18)	99.68		19			
29	C(19)	102.89		20			
30	C(20)	109.15		21			
31	C(21)	109.35		22			
32	C(22)	109.47		23			
33	C(23)	112.71		24			
34	C(24)	113.66		25			
35	C(25)	114.27		26			
36	C(26)	118.56		27			
37	C(27)	126.76		28			
38	C(28)	127.08		29			

	A	B	C	D	E	F	G
1	Subject	1					
2	Box	1					
3	StartDate	9/23/2024					
4	StartTime	7:43:36					
5	EndTime	8:28:59					
6	Session Duration (sec.)	1800.01					
7	Total Rewards Eaned	228					
8	Total L. Lever Presses	356					
9	L. Lever Presses Time Stamps			Lever Presses	Time in Minutes		
10	C(0)	21.17		1	0.352833333		
11	C(1)	37.58		2	0.626333333		
12	C(2)	47.16		3	0.786		
13	C(3)	49.84		4	0.830666667		
14	C(4)	50.15		5	0.835833333		
15	C(5)	50.26		6	0.837666667		
16	C(6)	55.33		7	0.922166667		
17	C(7)	58.11		8	0.9685		
18	C(8)	59.61		9	0.9935		
19	C(9)	60.11		10	1.001833333		
20	C(10)	60.86		11	1.014333333		
21	C(11)	63.44		12	1.057333333		
22	C(12)	74.15		13	1.235833333		
23	C(13)	84.09		14	1.4015		
24	C(14)	84.77		15	1.412833333		
25	C(15)	84.93		16	1.4155		
26	C(16)	96.01		17	1.600166667		
27	C(17)	96.29		18	1.604833333		
28	C(18)	99.68		19	1.661333333		
29	C(19)	102.89		20	1.714833333		
30	C(20)	109.15		21	1.819166667		
31	C(21)	109.35		22	1.8225		
32	C(22)	109.47		23	1.8245		
33	C(23)	112.71		24	1.8785		
34	C(24)	113.66		25	1.894333333		
35	C(25)	114.27		26	1.9045		
36	C(26)	118.56		27	1.976		
37	C(27)	126.76		28	2.112666667		
38	C(28)	127.08		29	2.118		

- Do the above instructions for each day of data. To make graphing easier, place all of the Time in Minutes and Lever Presses columns for every day on the same spreadsheet once you have cleaned up the data for each day.
 - When you copy the columns, right-click and look under the paste option, and choose the clipboard with number (i.e., values) to ensure the formulas do not transfer.



-

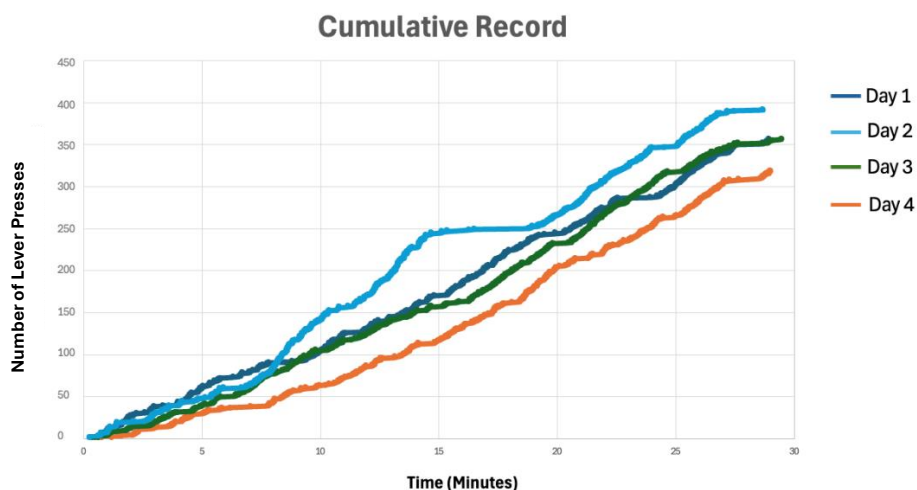
-
- The screenshot shows a menu titled 'More Scatter Charts...'. Under the 'Scatter' heading, there are five chart options: a standard scatter plot, a scatter plot with a trend line, a scatter plot with a non-linear trend line, a scatter plot with a step line, and a scatter plot with a smooth curve. The first option, the standard scatter plot, is highlighted with a red box. Below the 'Scatter' section, the 'Bubble' section is visible, showing two bubble chart options: one with three bubbles of different sizes and colors, and another with three bubbles of different sizes and shades of gray.

- 79 | Page

Day	Lever Presses	Time in Minutes
1	0.352833333	0.338
2	0.626333333	0.531
3	0.786	0.6015
4	0.830666667	0.608666667
5	0.835833333	1.446833333
6	0.837666667	1.531
7	0.922166667	1.6449
8		1.6449
12	1.057333333	1.659666667
13	1.235833333	1.670166667
14	1.4015	1.769833333
15	1.412833333	1.7765
16	1.4155	1.781166667
17	1.600166667	1.796166667
18	1.604833333	1.8715
19	1.661333333	2.142
20	1.714833333	2.201666667
21	1.819166667	2.2065
22	1.8225	2.2795
23	1.8245	2.288666667
24	1.8785	2.370833333
25	1.894333333	2.381666667
26	1.9045	2.457333333
27	1.976	2.466

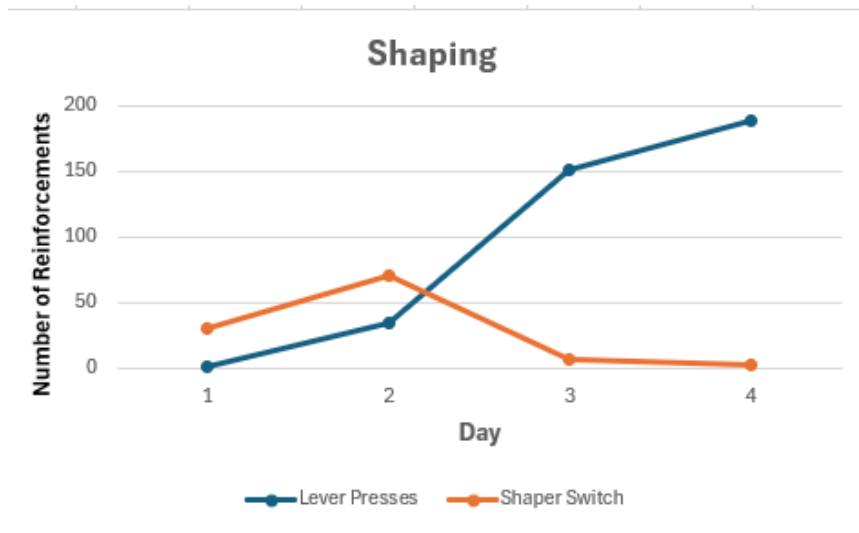
Delete then press button on right

- Do the same for every day of data in each series.
- Title the graph and both the horizontal and vertical axes.
- The final graph should look something like this:

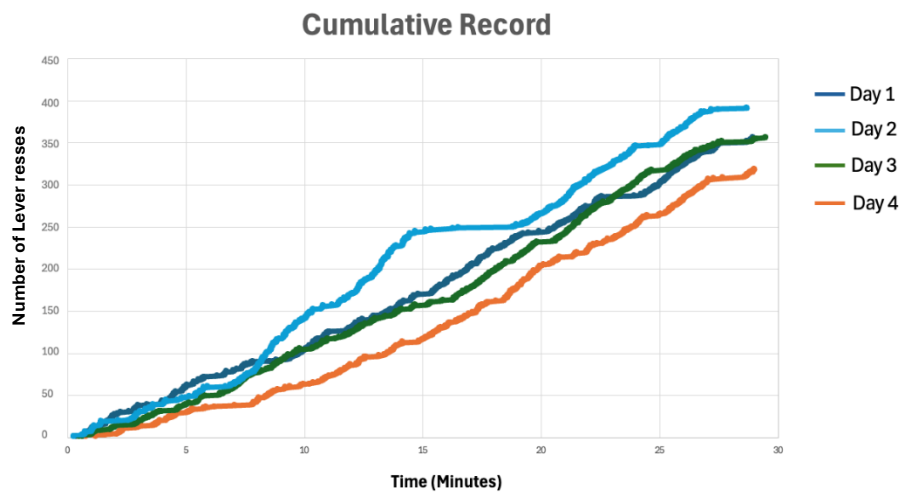


Appendix 3.3: Sample Graphs

Behavioral Trendline:



Cumulative Record:



Appendix 4.0: Rat Adoption Protocol

- 1.) The teaching assistant will keep a log with the rat identifiers and the students assigned to each rat before the beginning of the rat experiments.
- 2.) Potential adopter must express interest in adopting a rat to the teaching assistant.
- 3.) The teaching assistant will keep a log of potential adopters as well as the identified rat for potential adoption (Adopter Log). The Adopter Log will be emailed to IACUC@baylor.edu.
- 4.) Office of Research Compliance (ORC) will review the log, input any rat identifiers in the Adoption Form, and email the Adoption Form to the Attending Veterinarian (AV) for review.
- 5.) If there are multiple adoption requests for a rat, the adoption will be determined on a “first come, first served basis”. The other potential adopters will be asked if they would like to adopt a different rat that does not currently have an adopter.
- 6.) The AV will perform a physical exam of the rat and will certify that it is suitable for adoption in accordance with IACUC-13: Animal Disposition before it is eligible for release.
- 7.) The AV may, at his or her discretion, interview the potential adopter to determine the individual’s willingness and ability to provide for the animal’s welfare.
- 8.) Final approval of the adoption will be granted by the AV.
- 9.) The AV will email the completed Adoption Form to IACUC@baylor.edu.
- 10.) ORC will email the Adoption Form to the adopter for review and signature. After signing the Adoption Form, the adopter will email the signed Adoption Form to IACUC@baylor.edu.
- 11.) ORC will notify adopter, teaching assistant, and Dr. Riley of approval.
- 12.) ORC will keep a copy of the executed Adoption Form and email an executed copy to the adopter.
- 13.) The adopter will consult with the teaching assistant, or Dr. Riley, on a mutually agreed time to pick up the rat.

Appendix 5.0: Course Evaluation

It is important for us to receive your constructive feedback on your experiences in this course. Such evaluations are always important and they are the adult responsible and right thing to do. It is highly recommended that you complete each section as you engage in each experiment, but regardless, please turn in the document during your lab final exam.

Assignment 1: Handling, Magazine Training, and Shaping

1. What did you like best about this experiment?
2. Did you encounter issues while running this experiment?
3. What could be improved about this experiment?
4. This experiment helped me to better understand the concepts outlined in the lecture and textbook.

Yes No (circle one)

Assignment 2: Continuous Reinforcement, Extinction and Recovery

1. What did you like best about this experiment?
2. Did you encounter issues while running this experiment?
3. What could be improved about this experiment?
4. This experiment helped me to better understand the concepts outlined in the lecture and textbook.

Yes No (circle one)

Assignment 3: Partial Reinforcement and Partial Reinforcement Extinction Effect

1. What did you like best about this experiment?
2. Did you encounter issues while running this experiment?
3. What could be improved about this experiment?

4. This experiment helped me to better understand the concepts outlined in the lecture and textbook.
- Yes No (circle one)

Assignment 4: Choice Behavior

1. What did you like best about this experiment?
 2. Did you encounter issues while running this experiment?
 3. What could be improved about this experiment?
 4. This experiment helped me to better understand the concepts outlined in the lecture and textbook.
- Yes No (circle one)

Assignment 5: Stimulus Discrimination and Stimulus Discrimination Reversal

1. What did you like best about this experiment?
 2. Did you encounter issues while running this experiment?
 3. What could be improved about this experiment?
 4. This experiment helped me to better understand the concepts outlined in the lecture and textbook.
- Yes No (circle one)

Rat Olympics

1. What did you like best about this activity?
2. Did you encounter issues while running this experiment?
3. What could be improved about this activity?

4. Should the Rat Olympics be included in the lab next semester?

Yes No (circle one)

General Issues about the Learning and Behavior Lab:

1. What was your overall impression of this lab?
2. What improvements should we make to the overall lab structure?
3. What is your opinion of Sniffy? In what ways was it beneficial?
4. Any additional comments you would like to make?

Record of Rat's Weight and Water Intake

[illegible]

80% value of rat’s initial weight _____